

BERT in IR

The turning point!

- **2018** was a **turning point** for Natural Language Processing
 - BERT
 - OpenAI GPT
 - ELMO
 - ULM fit



BERT: Bidirectional Encoder Representations from Transformers

Language model: probability distribution over sequences of words

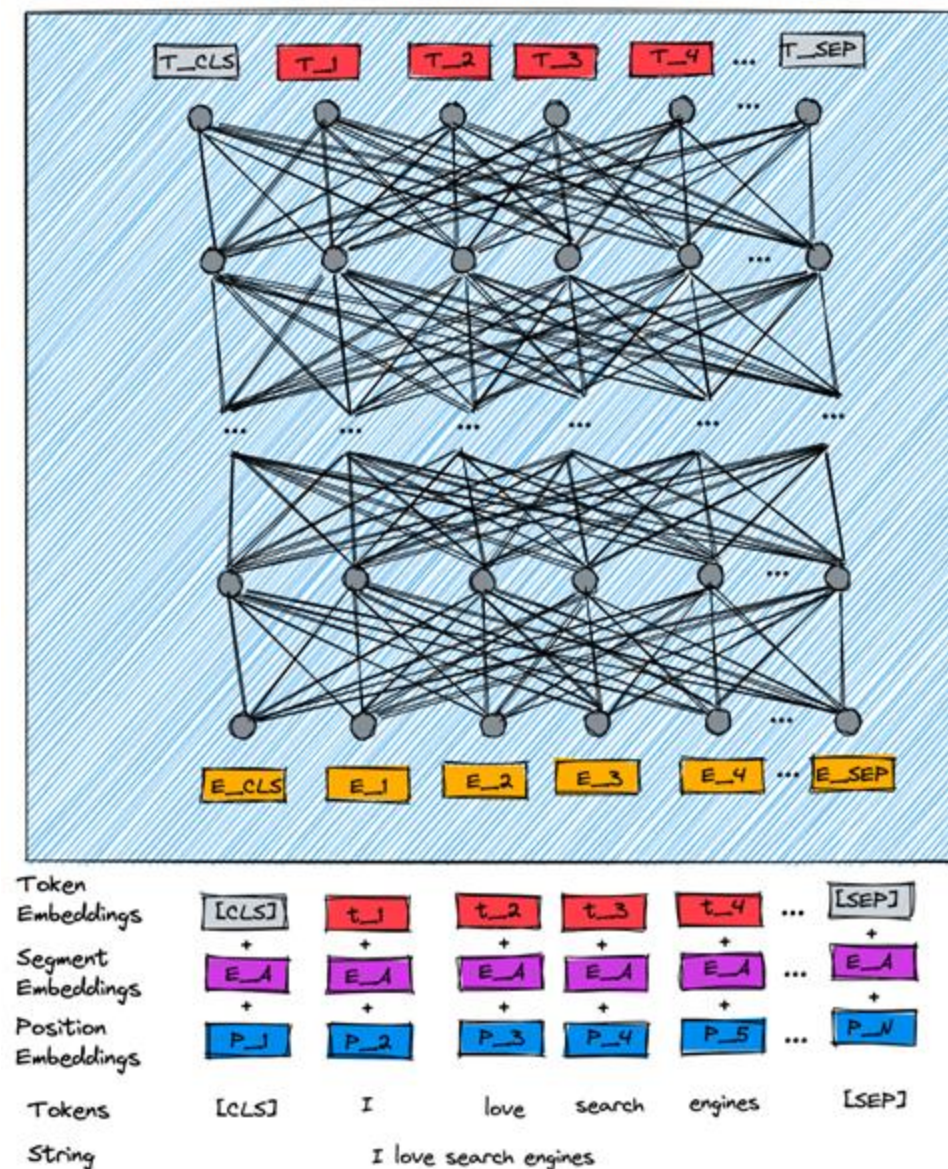
- context of a word based on its surroundings
- bidirectional context: left and right

BERT is trained on two tasks:

- Masked Language Model
- Next Sentence Prediction

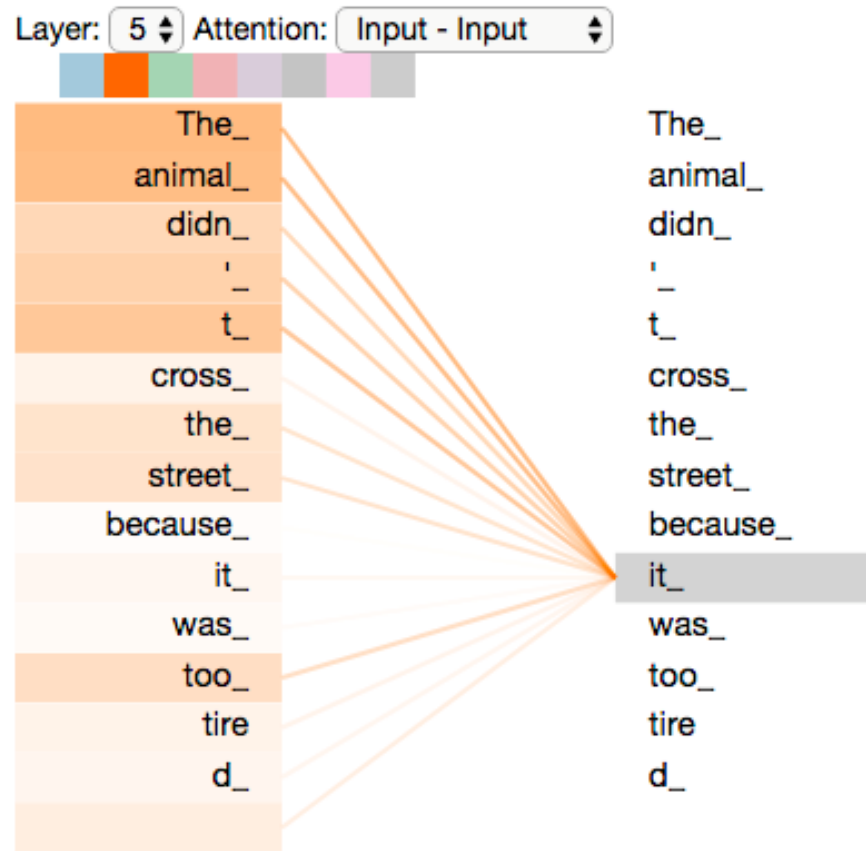
Special tokens:

- **[CLS]** Classification token, used as pooling operator to get a single vector per sequence
- **[MASK]** Used in the masked language model, to predict this word
- **[SEP]** Used to separate sentences



Self-Attention

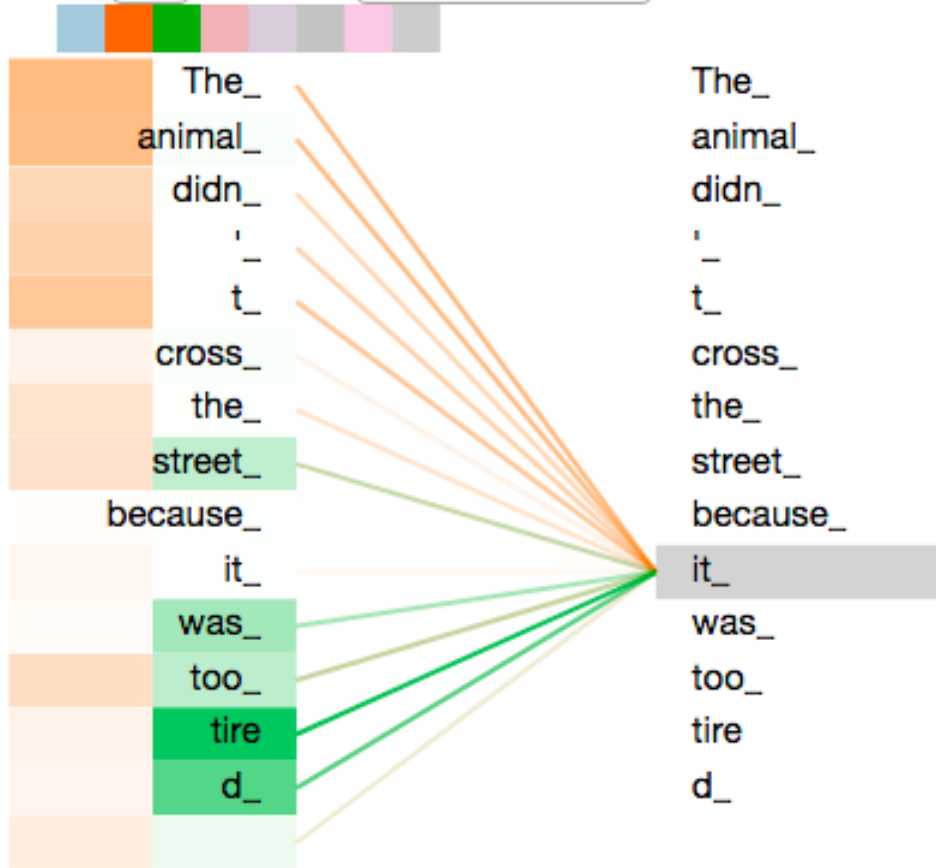
- The basic unit of a Transformer is the Self-Attention mechanism
 - Ability to learn which bits of a sequence are most important for a given task



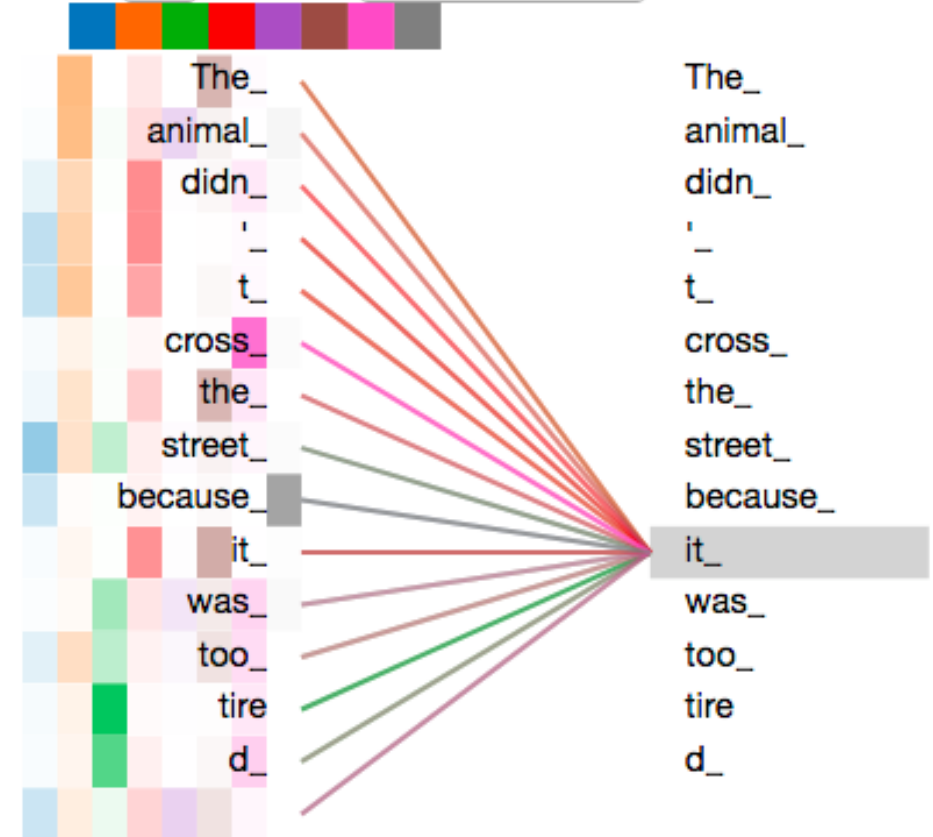
Multi-headed Attention

The single attention head can be replicated to capture different contextual relationship

Layer: 5 Attention: Input - Input



Layer: 5 Attention: Input - Input



BERT Training and Tuning to ad-hoc tasks

1 - **Semi-supervised** training on large amounts of text (books, wikipedia..etc).

The model is trained on a certain task that enables it to grasp patterns in language. By the end of the training process, BERT has language-processing abilities capable of empowering many models we later need to build and train in a supervised way.

Semi-supervised Learning Step

Model:



Dataset:



Objective:

Predict the masked word
(language modeling)

2 - **Supervised** training on a specific task with a labeled dataset.

Supervised Learning Step

Model:
(pre-trained
in step #1)



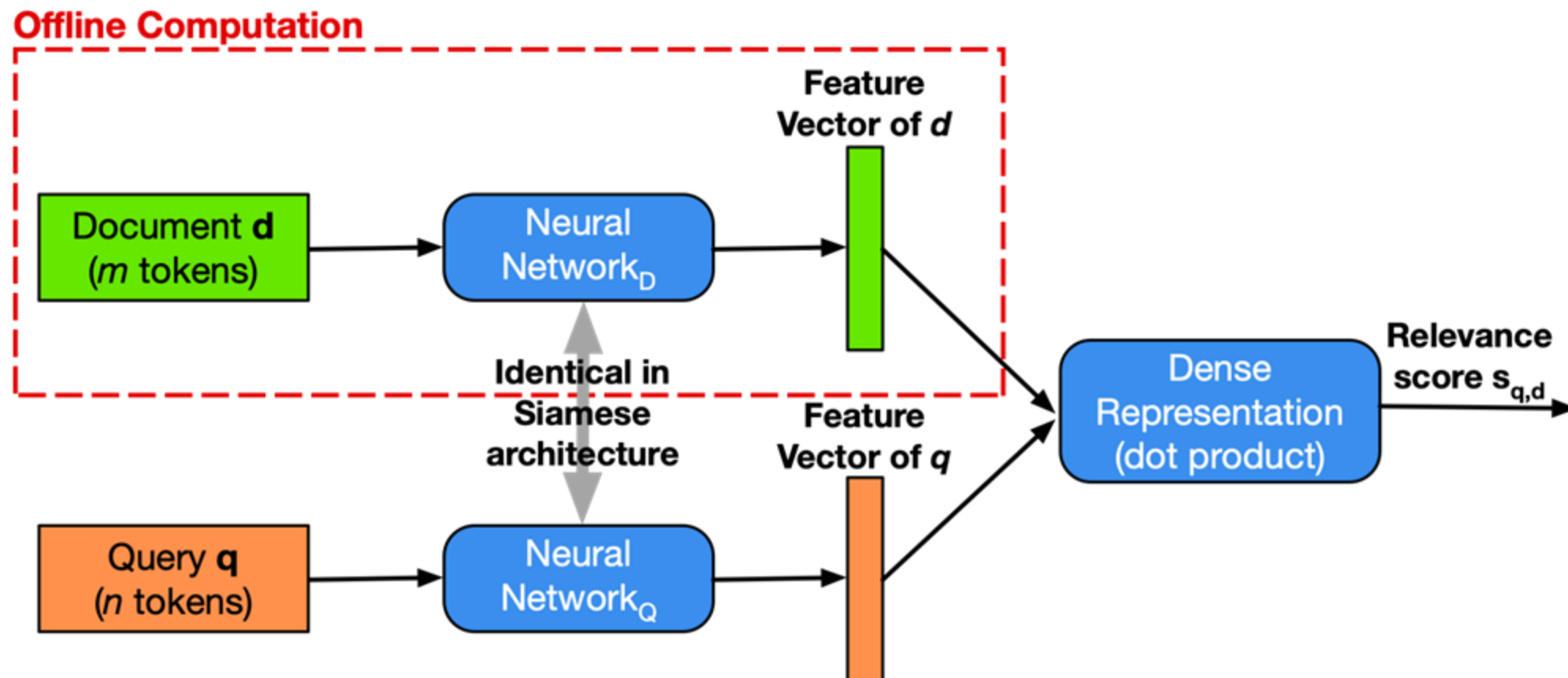
75% Spam
25% Not Spam



Dataset:

Email message	Class
Buy these pills	Spam
Win cash prizes	Spam
Dear Mr. Atreides, please find attached...	Not Spam

Neural Approaches for IR

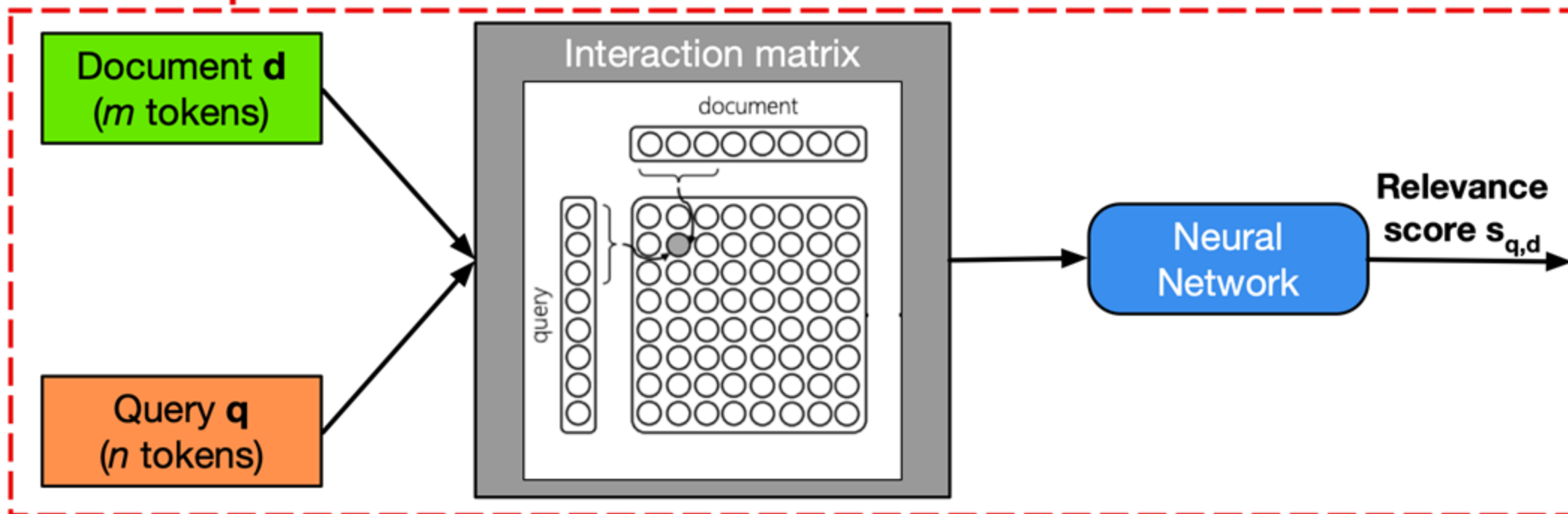


- **Representation-based methods**

- Each document and a query are separately represented as dense vectors through a sequence of neural computations (**offline** for the documents, **online** for the query)
- The final ranking is based on the similarity of the two representative vectors

Neural Approaches for IR

Online Computation



- **Interaction-based methods**

- The word or term-level similarity (interaction matrix) of a query and a document is explored **online** first based on their embedding vectors
- The final ranking is obtained by an additional sequence of neural computation

Interaction-based Methods

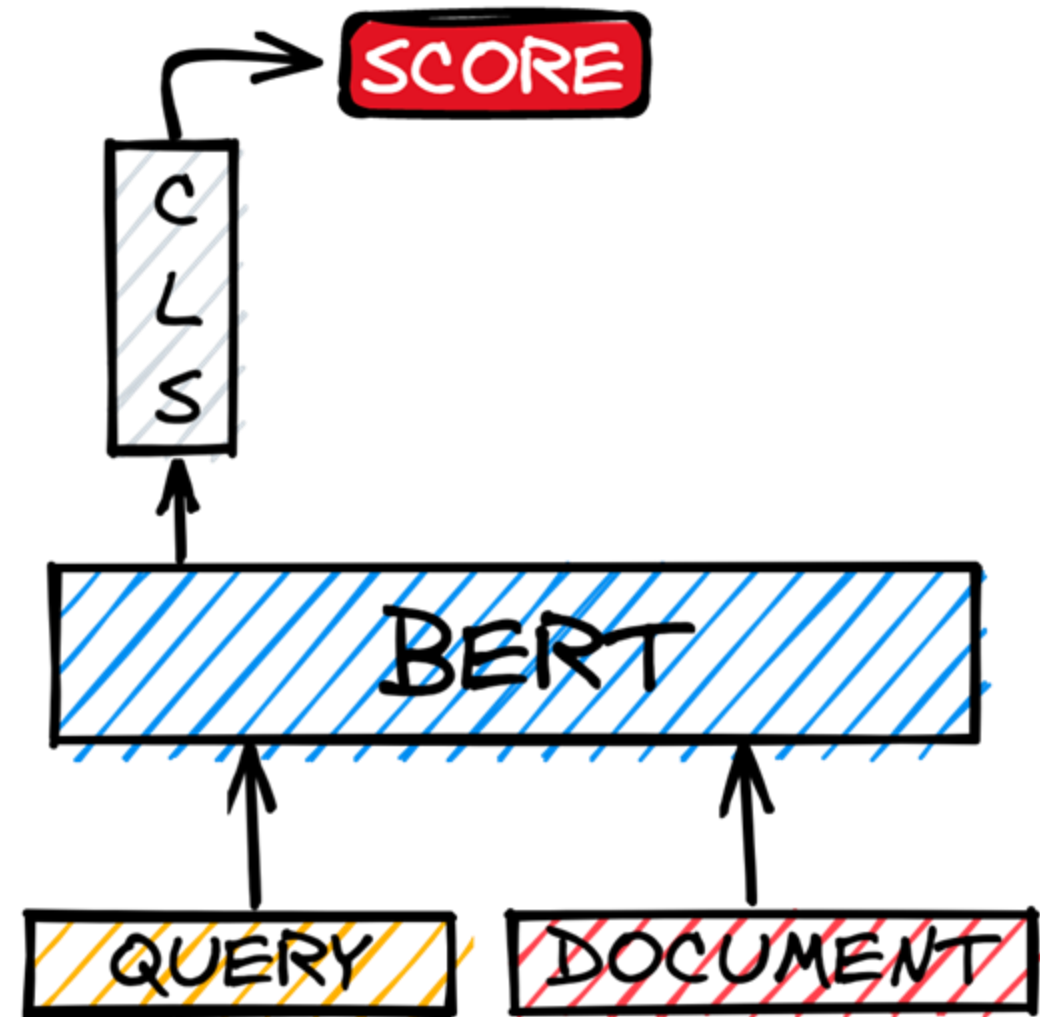
MonoBERT (vanilla BERT)

Idea: query and document are jointly cross-encoded, and the model output tuned to provide a ranking score

Input : **[CLS]** q **[SEP]** d **[SEP]**

Output: **[CLS]** score

The contextual embedding of the [CLS] token is the aggregate relevance score of the query-document pair, to be directly used for ranking



Fine-Tuning MonoBERT for ranking

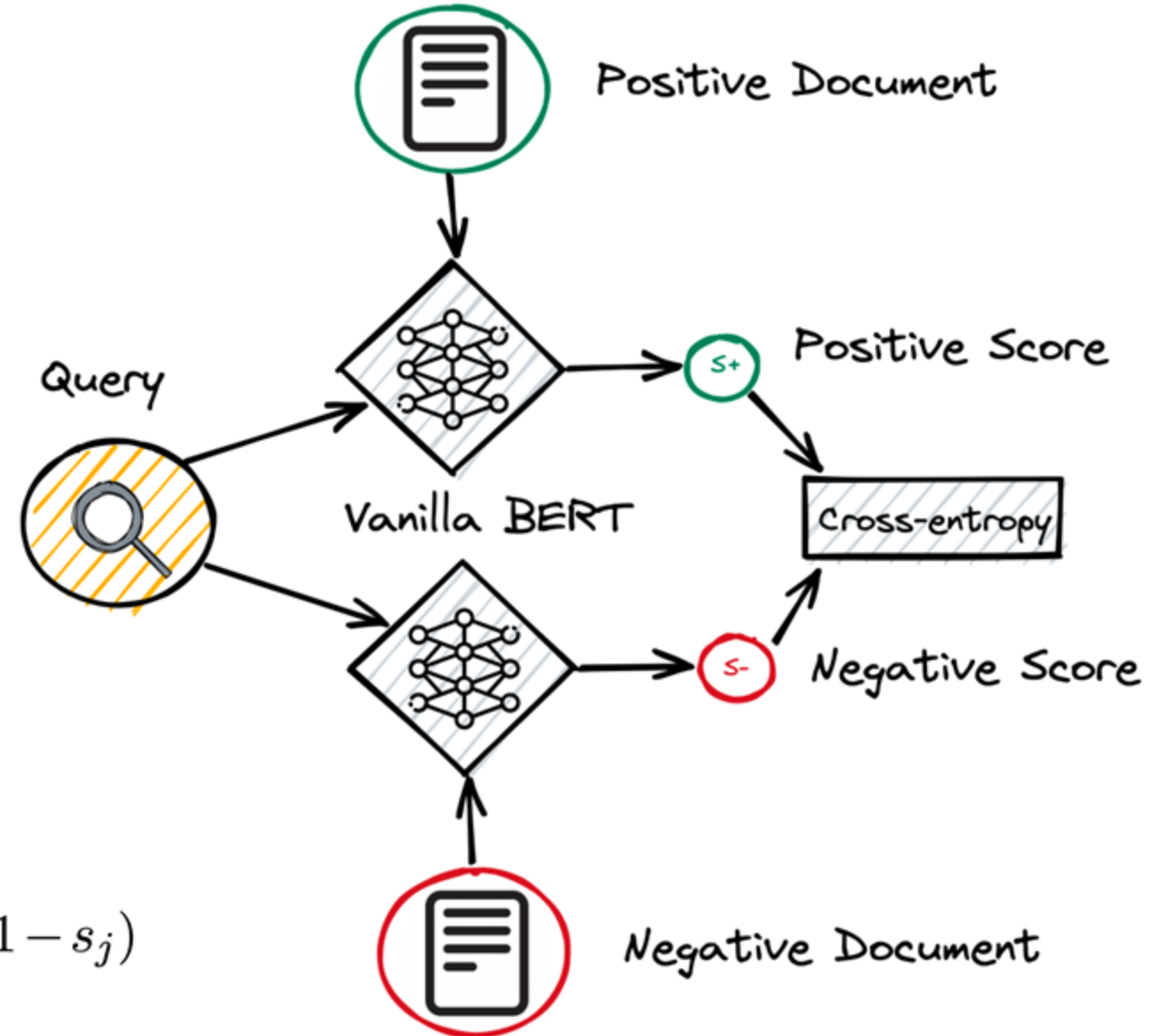
Vanilla BERT is trained on two tasks:

- Masked Language Model
- Next Sentence Prediction

Adapting the model to a ranking scenario leveraging to fine-tuning:

- Point-wise model
- Pair-wise training approach
 - Cross-entropy loss

$$L_{\text{mono}} = - \sum_{j \in J_{\text{pos}}} \log(s_j) - \sum_{j \in J_{\text{neg}}} \log(1 - s_j)$$



MonoBERT Performance

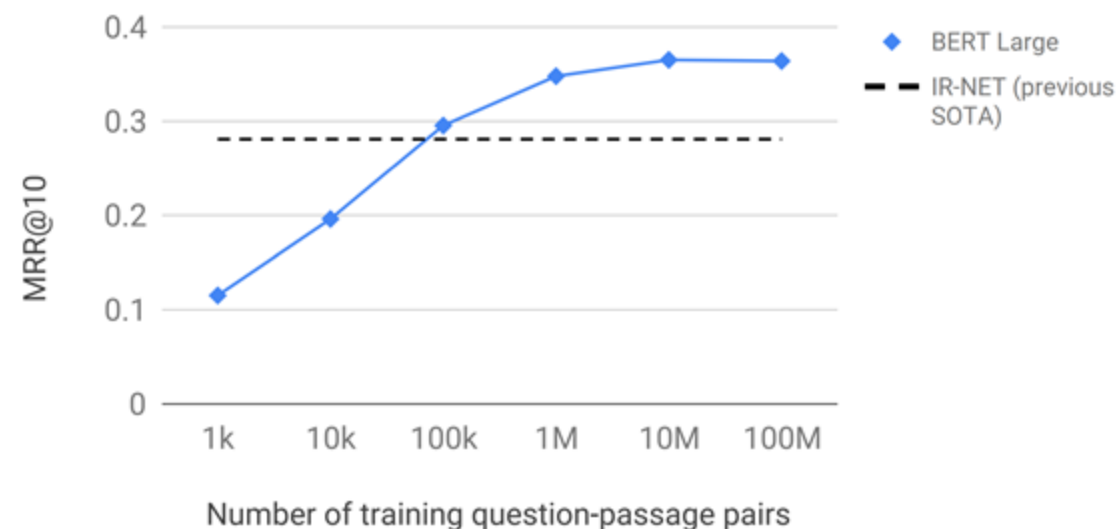
A general purpose language model,
fine-tuned only on the specific task,
outperforms SOTA solutions

100k training pairs only (0,3% of the
full training data)

- 10k queries
- 10 passages per query by BM25

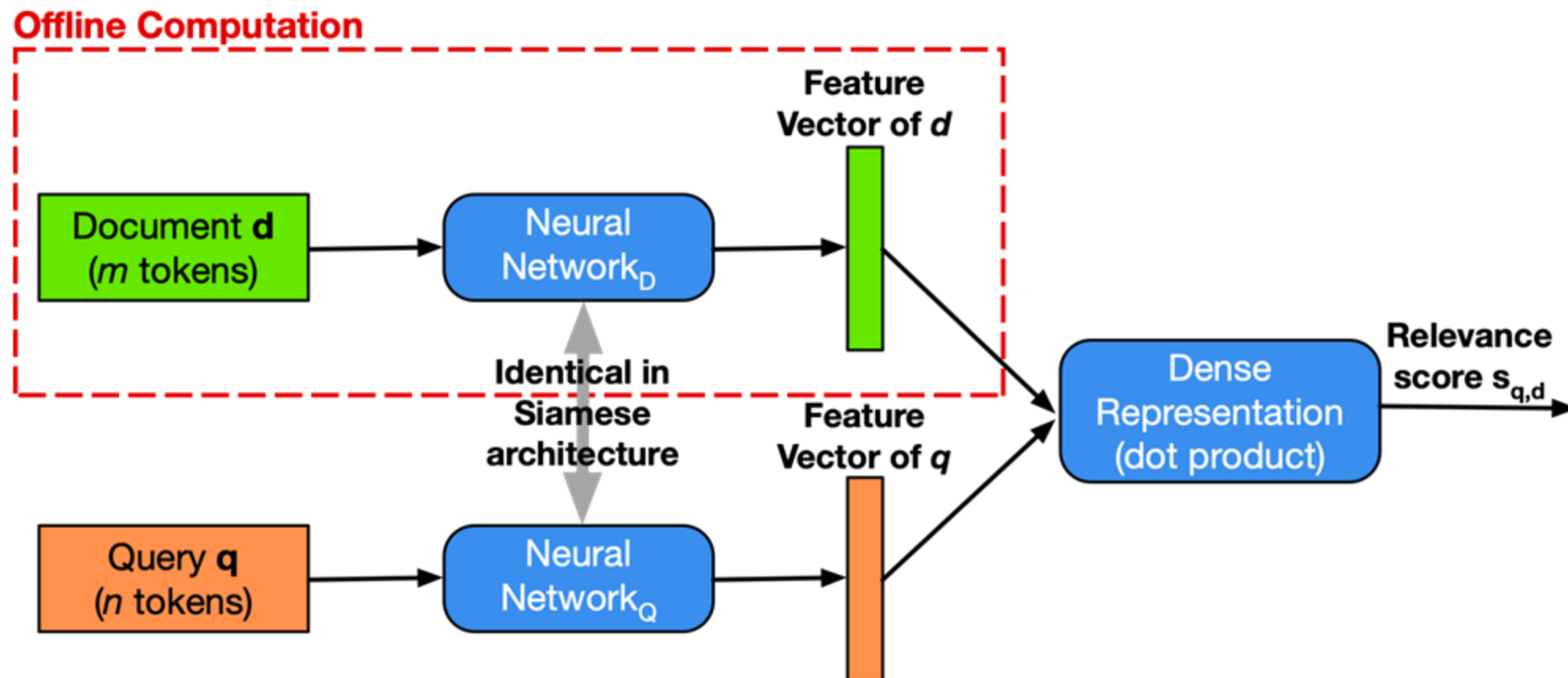
and BERT is already better than
SOTA!!!

Method	MS MARCO MRR@10		TREC-CAR MAP
	Dev	Eval	Test
BM25 (Lucene, no tuning)	16.7	16.5	12.3
BM25 (Anserini, tuned)	-	-	15.3
Co-PACRR* (MacAvaney et al., 2017)	-	-	14.8
KNRM (Xiong et al., 2017)	21.8	19.8	-
Conv-KNRM (Dai et al., 2018)	29.0	27.1	-
IRNet [†]	27.8	28.1	-
BERT Base	34.7	-	31.0
BERT Large	36.5	35.8	33.5



Representation-based Methods

Neural Approaches for IR



- **Representation-based methods**

- Each document and a query are separately represented as dense vectors through a sequence of neural computations (**offline** for the documents, **online** for the query)
- The final ranking is based on the similarity of the two representative vectors

Neural Approaches for IR (from 2019)

- **Representation-based methods**

- Dense retrieval has many desired properties

- Fully-learnable representation
 - Easy integration with fine-tuning
 - Efficient: support from approximate nearest neighbor (ANN) search

- Representation-based methods overcome limitations of sparse retrieval

- vocabulary mismatch

Dense Representations

Dense representations

- Learn a projection of feature vectors into a *low-dimensional* space $\mathbb{R}^k$, $k \ll |F|$, of continuous space word representations (embedding).
- **Word2Vec** [MCC13]
 - Two approaches (CBOW, Skip-gram) for learning continuous vector representations of words.
 - CBOW: predicts a word from its context
 - Skip-gram: predicts a context from a word
 - Both implemented as a two-layers linear neural network
 - input and output words one-hot representations are encoded/decoded into/from a dense representation of smaller dimensionality
 - No need for human-labeled data!
 - Context window size: longer windows capture more semantic, less syntax

Dense Representations

More recent approaches

- **FastText**: extends the W2V embedding model with *ngrams* of a word
 - Embedding of a word: sum of its embedding and of the embedding of its ngrams.
- **Doc2Vec** [ML14]:
 - Doc2Vec extends Word2Vec by adding input dimensions for identifiers of documents
 - Documents ids are projected in the same space of words
 - Document embeddings from the words composing them
 - Document embeddings are **vectorial representations of documents**
 - Can be used for any task
- **BERT** [D19]
 - Contextualized embeddings, i.e., the embedding of a word depends from its context

Dense Representations for IR

Several contributions in the last years

- ColBERT [KZ20]
- ANCE [XXL21]
- STAR / ADORE [ZML21]

Single-representation vs. Multiple-representation methods

- Single representation: one embedding for each document (ANCE)
- Multiple representation: one embedding for each term of the document (ColBERT)

[KZ20] Khattab and Zaharia. **ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT**. ACM SIGIR 2020.

[XXL21] Xiong *et al.* **Approximate Nearest Neighbor Negative Contrastive Learning for Dense Text Retrieval**. ICLR 2021.

[ZML21] Zhan *et al.* **Optimizing Dense Retrieval Model Training with Hard Negatives**. ACM SIGIR 2021.

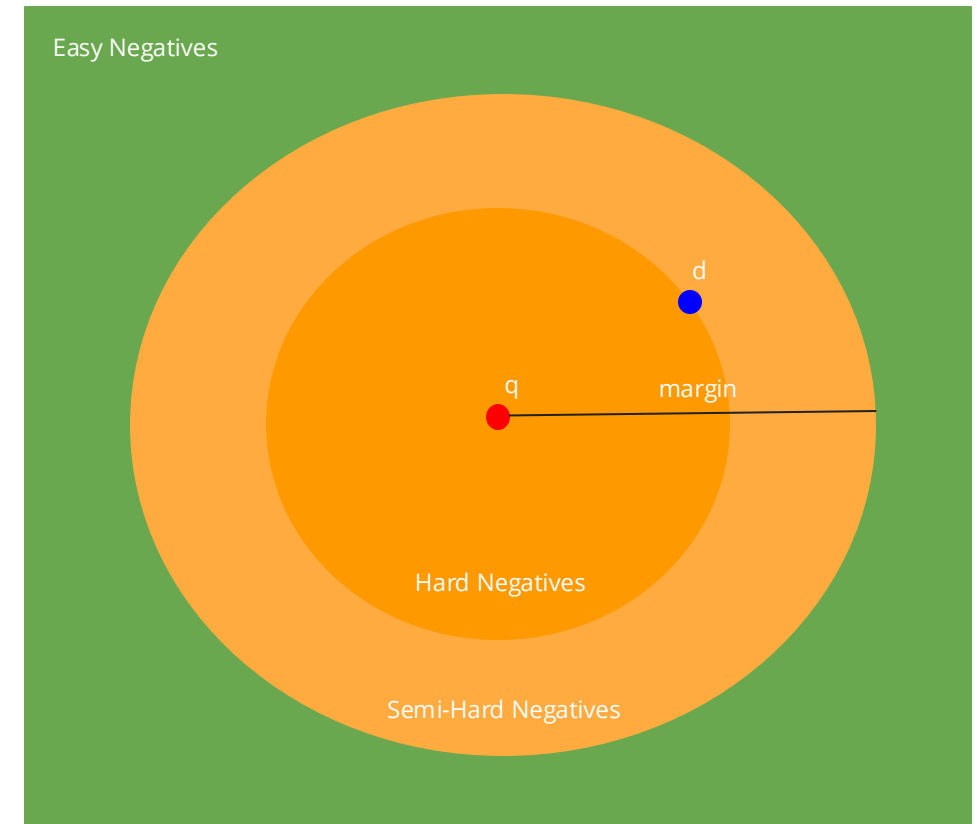
Learning Dense Representations for IR

- Pairs or triples of data per input sample (q, d^+, d^-)
 - Get an embedding for each input sample
- Define a metric function to measure the similarity between embeddings
 - Cosine similarity
 - Dot product
- Train the model to produce
 - close embeddings for the two inputs, in case the inputs are similar
 - distant embeddings for the two inputs, in case they are dissimilar
- Losses capture relative distances between input samples

Losses and Negative Selection

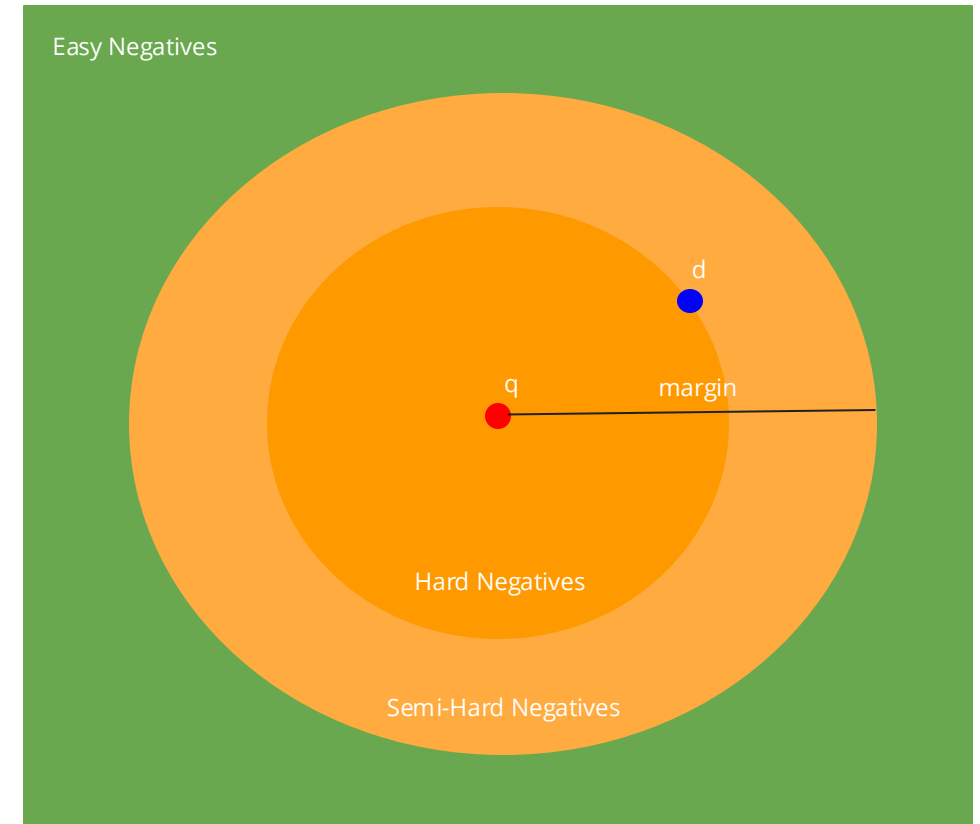
- Pairwise/Triplet loss
 - distance between q and d^+ should be 0
 - distance between q and d^- should be $> m$
- Positive docs are known
 - from supervision
- Negative docs
 - How to choose them?
- Several strategies
 - In-batch selection
 - Cross-batch selection
 - Random sampling

$$L(q, d) = \begin{cases} d(q, d) & \text{if } d \in D^+ \\ \max(0, m - d(q, d)) & \text{if } d \in D^- \end{cases}$$

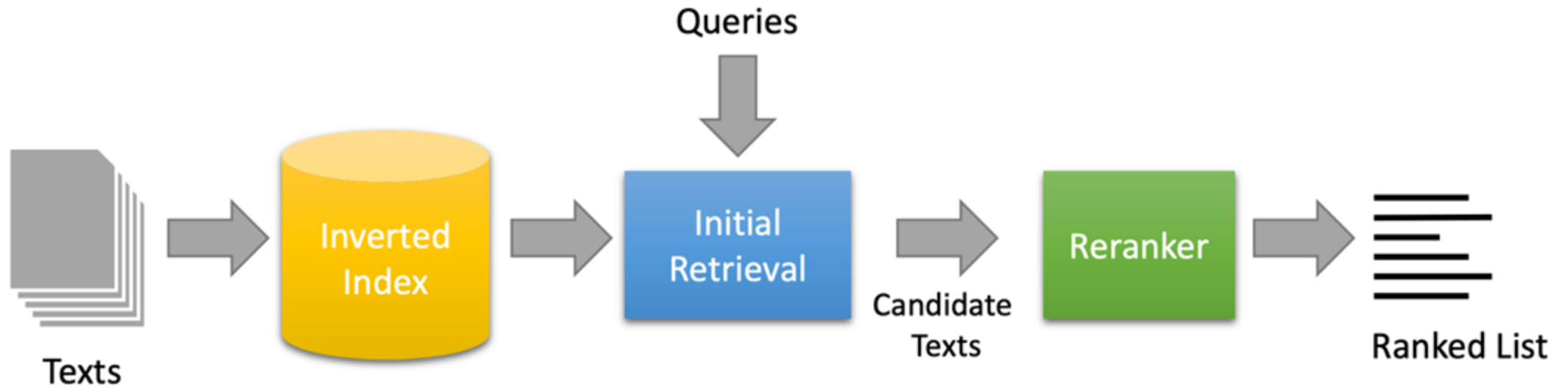


Losses and Negative Selection

- What about Hard Negatives (HN)?
 - the most important ones!
- Static Sampling vs. Dynamic Sampling
- Static Sampling
 - Hard negatives are pre-computed before training and **never change**
- Dynamic Sampling
 - Hard negative are the current top-ranked irrelevant documents produced by the dense model under training
 - **computed at each training step**

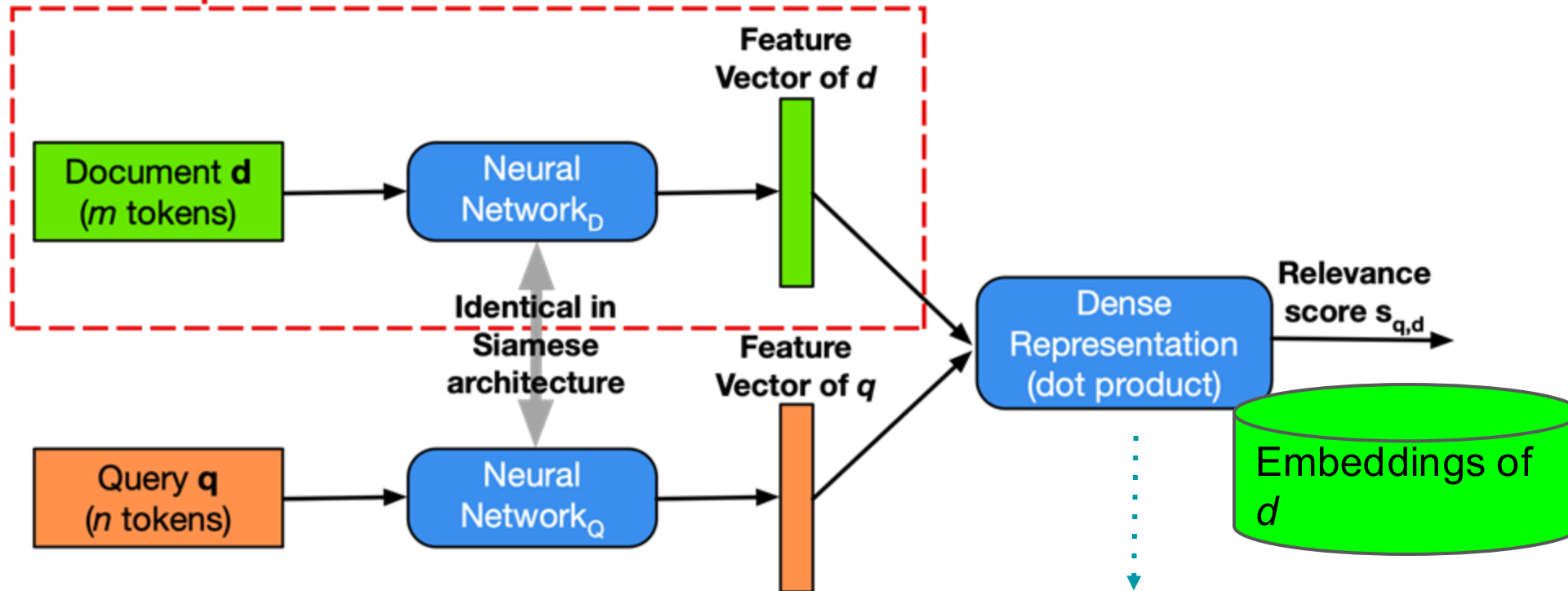


How Do Search Engines Work?



Dense Representations and Retrieval

Offline Computation

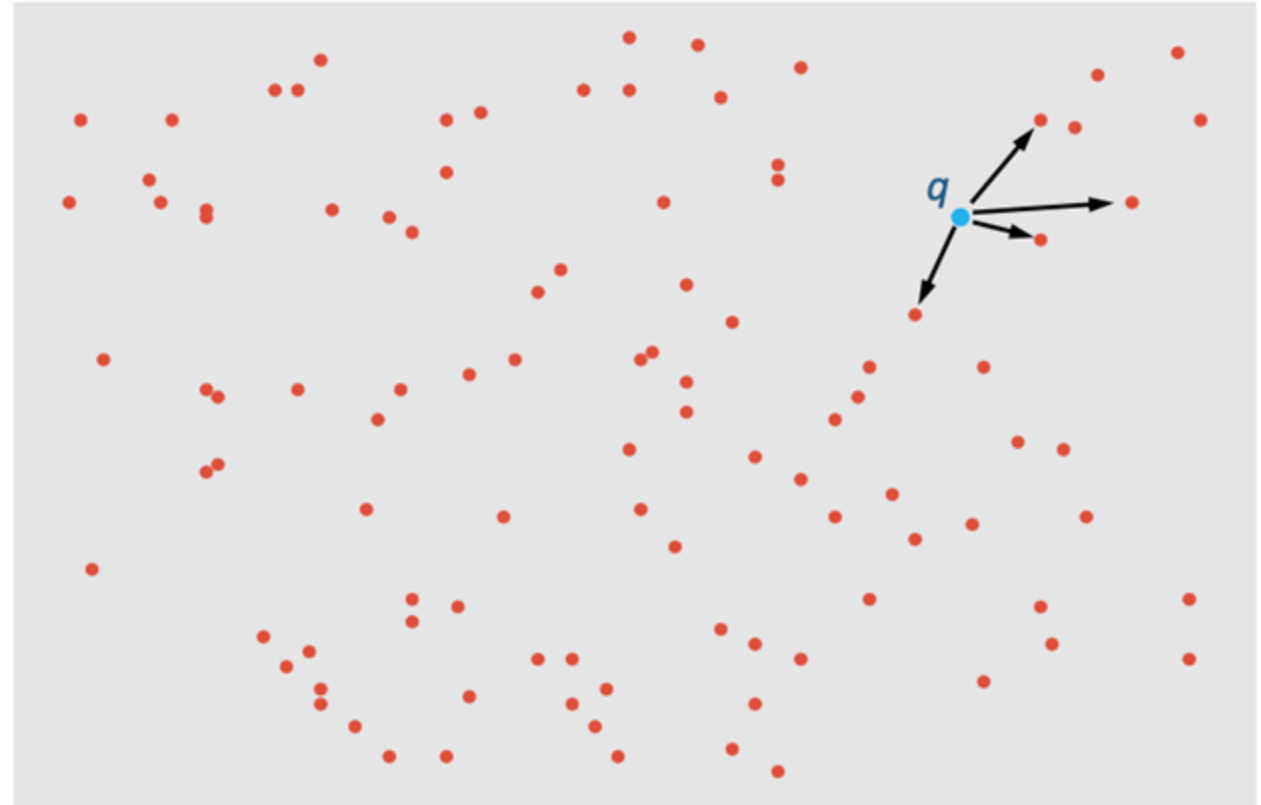


k Nearest Neighbour (kNN) Retrieval

- Closest k points in the N-dimensional space
- Exact vs. Approximate Nearest Neighbour
- Several techniques targeting efficiency

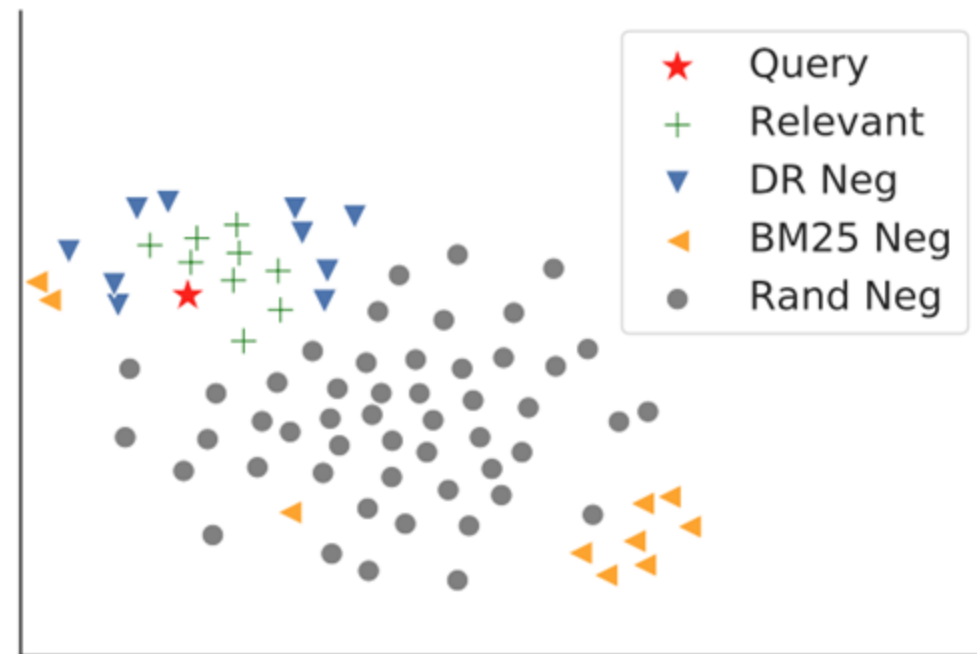
k Nearest Neighbour

- Retrieval as finding the most similar docs with the query
 - dot product
 - cosine similarity
- $O(n \log k)$
 - All d 's against q
 - Heap data structure that keeps k elements with higher similarity
- Slow/Exact vs Fast/Approximate
- Several techniques for ANN
 - Trees/Graphs/LSH/Quantization

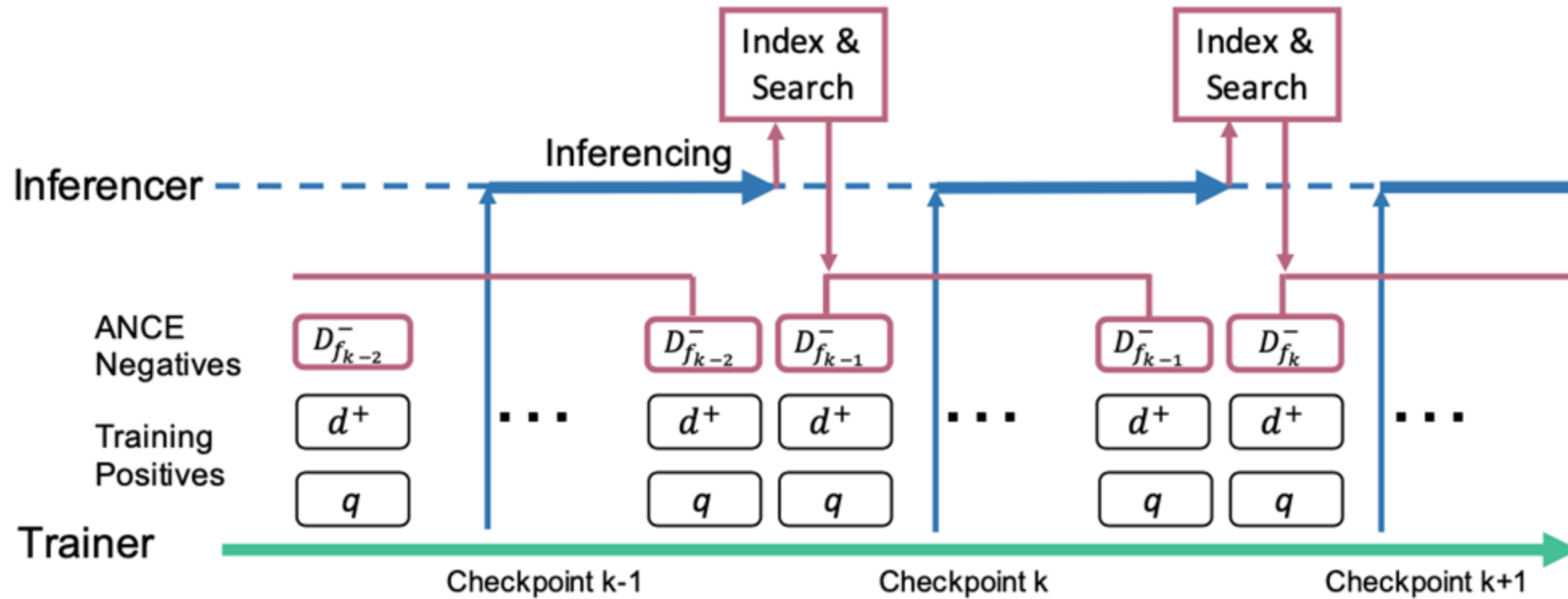


ANCE

- Sampling of HN is very important
- Dynamic hard negatives (HN)
 - Model learned so far used to select HN
- BERT Siamese/Dual Encoder
 - Shared between q and d
 - Single-representation technique
 - One vector per document
- Exact kNN retrieval



ANCE

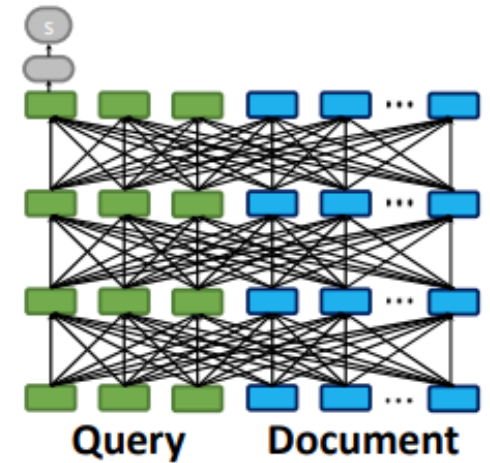


	MARCO Dev Passage Retrieval		TREC DL Passage NDCG@10		TREC DL Document NDCG@10	
	MRR@10	Recall@1k	Rerank	Retrieval	Rerank	Retrieval
Sparse & Cascade IR						
BM25	0.240	0.814	–	0.506	–	0.519
Best DeepCT	0.243	n.a.	–	n.a.	–	0.554
Best TREC Trad Retrieval	0.240	n.a.	–	0.554	–	0.549
BERT Reranker	–	–	0.742	–	0.646	–
Dense Retrieval						
Rand Neg	0.261	0.949	0.605	0.552	0.615	0.543
NCE Neg	0.256	0.943	0.602	0.539	0.618	0.542
BM25 Neg	0.299	0.928	0.664	0.591	0.626	0.529
DPR (BM25 + Rand Neg)	0.311	0.952	0.653	0.600	0.629	0.557
BM25 → Rand	0.280	0.948	0.609	0.576	0.637	0.566
BM25 → NCE Neg	0.279	0.942	0.608	0.571	0.638	0.564
BM25 → BM25 + Rand	0.306	0.939	0.648	0.591	0.626	0.540
ANCE (FirstP)	0.330	0.959	0.677	0.648	0.641	0.615
ANCE (MaxP)	–	–	–	–	0.671	0.628

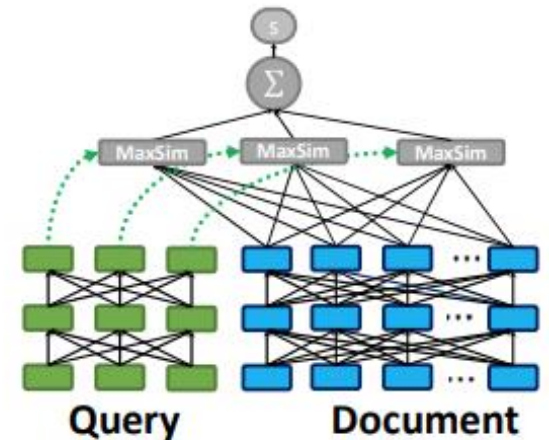
ColBERT: Contextualized Late Interaction over BERT

- Late interaction as a way to fight computational burden
- Two encoders: query and document encoder
 - end-to-end differentiable
- Term-based representation of documents
 - Each document is a set of vectors (one for each term)
 - They can be precomputed offline
- Queries are unknown
 - Contextualized representations are computed online
- Similarity between Q and D
- Approximate kNN retrieval

$$S_{q,d} := \sum_{i \in [|E_q|]} \max_{j \in [|E_d|]} E_{q_i} \cdot E_{d_j}^T$$



(c) All-to-all Interaction
(e.g., BERT)



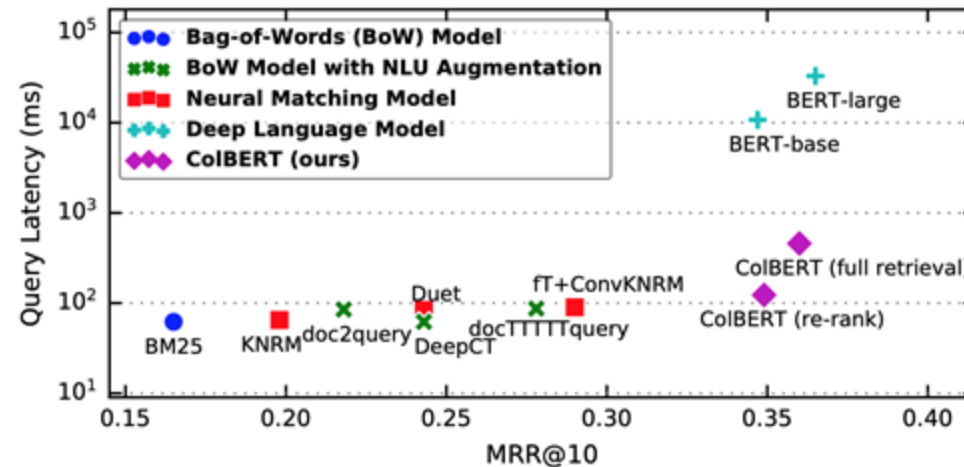
(d) Late Interaction
(i.e., the proposed ColBERT)

ColBERT: Contextualized Late Interaction over BERT

Method	MRR@10 (Dev)	MRR@10 (Eval)	Re-ranking Latency (ms)	FLOPs/query
BM25 (official)	16.7	16.5	-	-
KNRM	19.8	19.8	3	592M (0.085×)
Duet	24.3	24.5	22	159B (23×)
fastText+ConvKNRM	29.0	27.7	28	78B (11×)
BERT _{base} [25]	34.7	-	10,700	97T (13,900×)
BERT _{base} (our training)	36.0	-	10,700	97T (13,900×)
BERT _{large} [25]	36.5	35.9	32,900	340T (48,600×)
ColBERT (over BERT _{base})	34.9	34.9	61	7B (1×)

ColBERT: Contextualized Late Interaction over BERT

Method	MRR@10 (Dev)	MRR@10 (Local Eval)	Latency (ms)	Recall@50	Recall@200	Recall@1000
BM25 (official)	16.7	-	-	-	-	81.4
BM25 (Anserini)	18.7	19.5	62	59.2	73.8	85.7
doc2query	21.5	22.8	85	64.4	77.9	89.1
DeepCT	24.3	-	62 (est.)	69 [2]	82 [2]	91 [2]
docTTTTTquery	27.7	28.4	87	75.6	86.9	94.7
ColBERT _{L2} (re-rank)	34.8	36.4	-	75.3	80.5	81.4
ColBERT _{L2} (end-to-end)	36.0	36.7	458	82.9	92.3	96.8

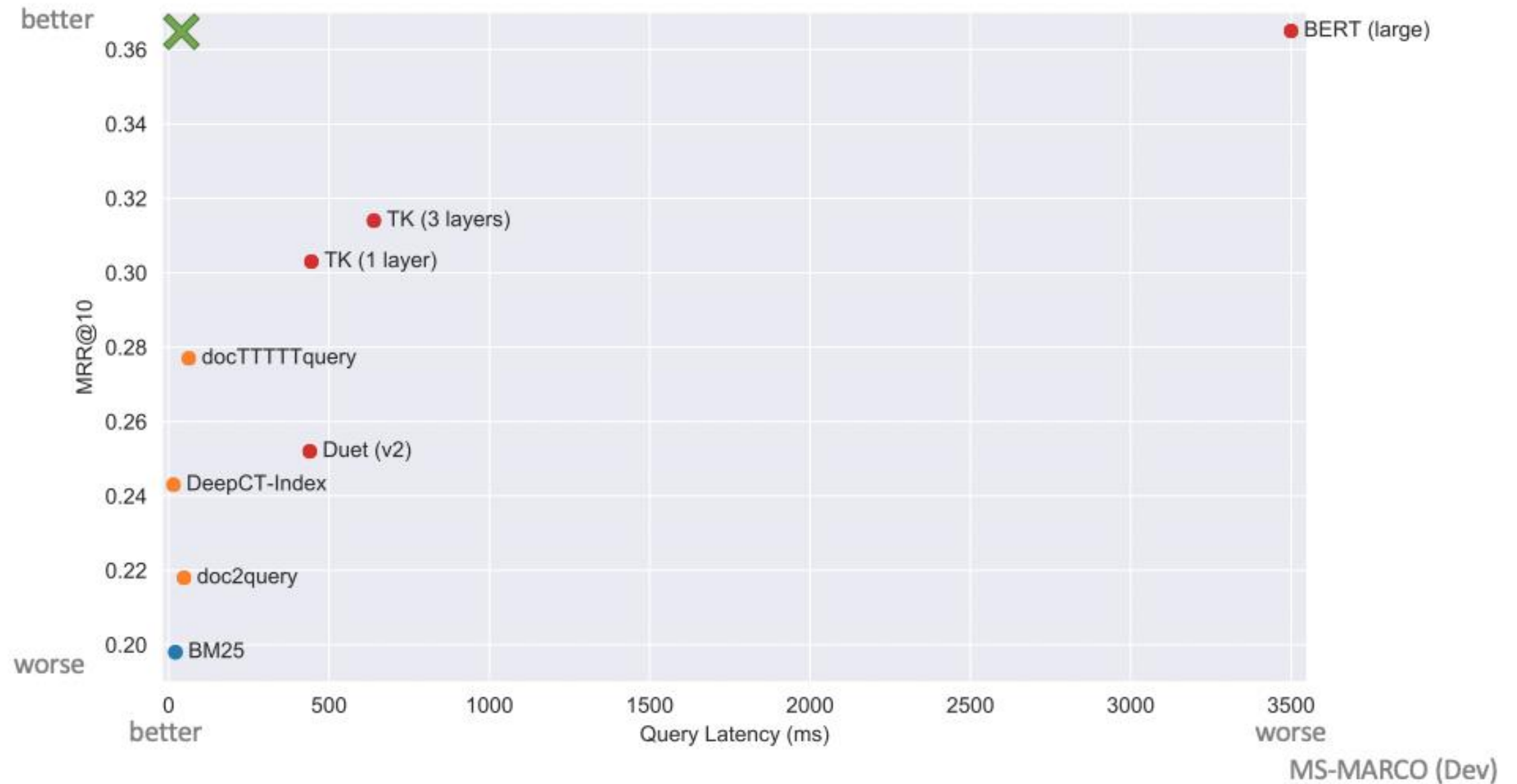


Efficiency Matters!

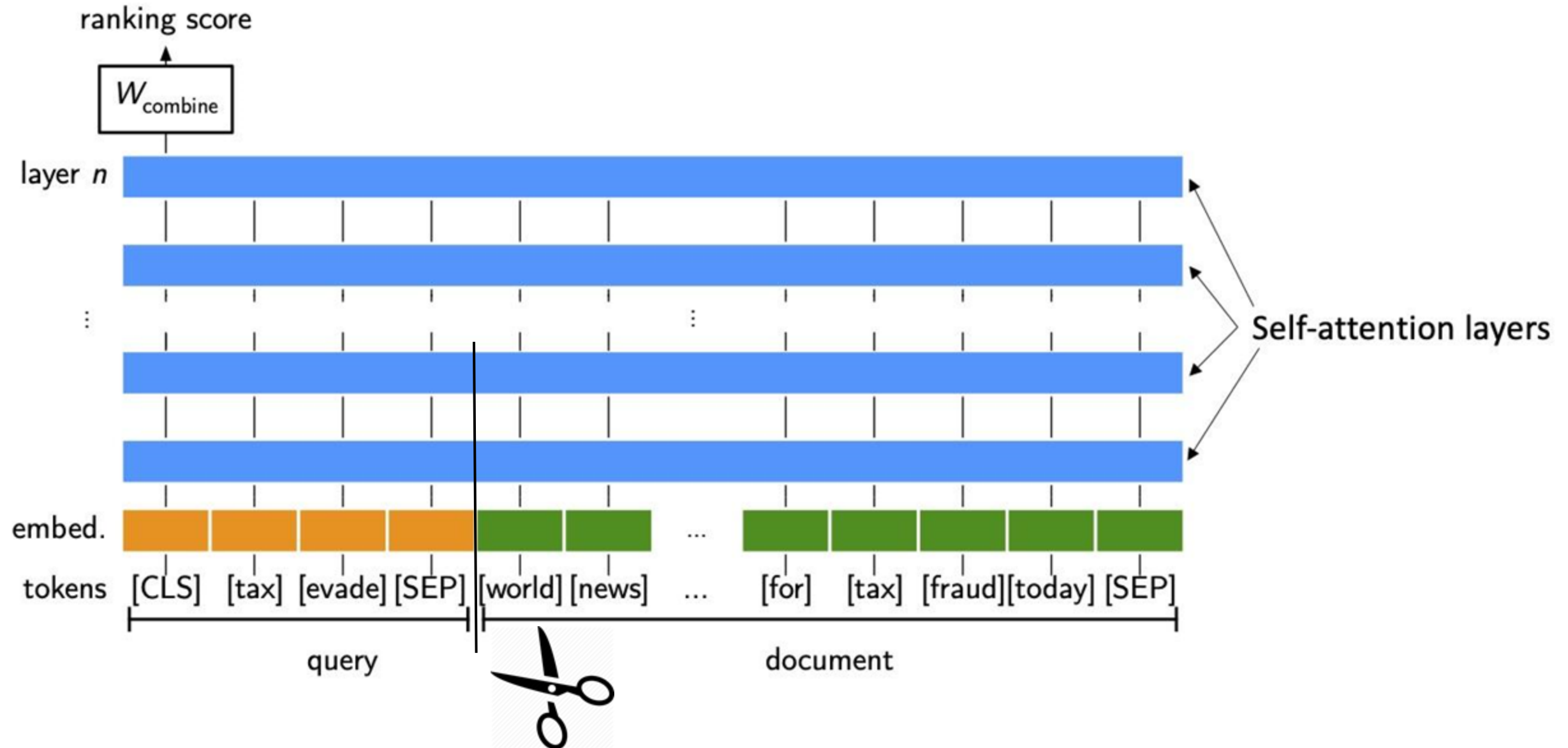
Efficiency matters!!!

- Deep Transformer Networks outperform SOTA in many NLP and IR tasks
- These models are huge in size, thus expensive to run
 - In one year pretrained transformer models grew from about 110 million parameters (GPT) to over 8.3 billion (Megatron-LM) or 175 billion (GPT-3).
- Application to Web Search Engines (IR)
 - Ad-hoc Retrieval: find the top- k most relevant documents answering a query in 100 msec.
 - Substantial impact on the query processing performance
- Several proposals targeting efficiency with neural networks in IR
 - De-coupled Interaction
 - Use of Knowledge Distillation, Quantization, Early exit

BERT for document ranking

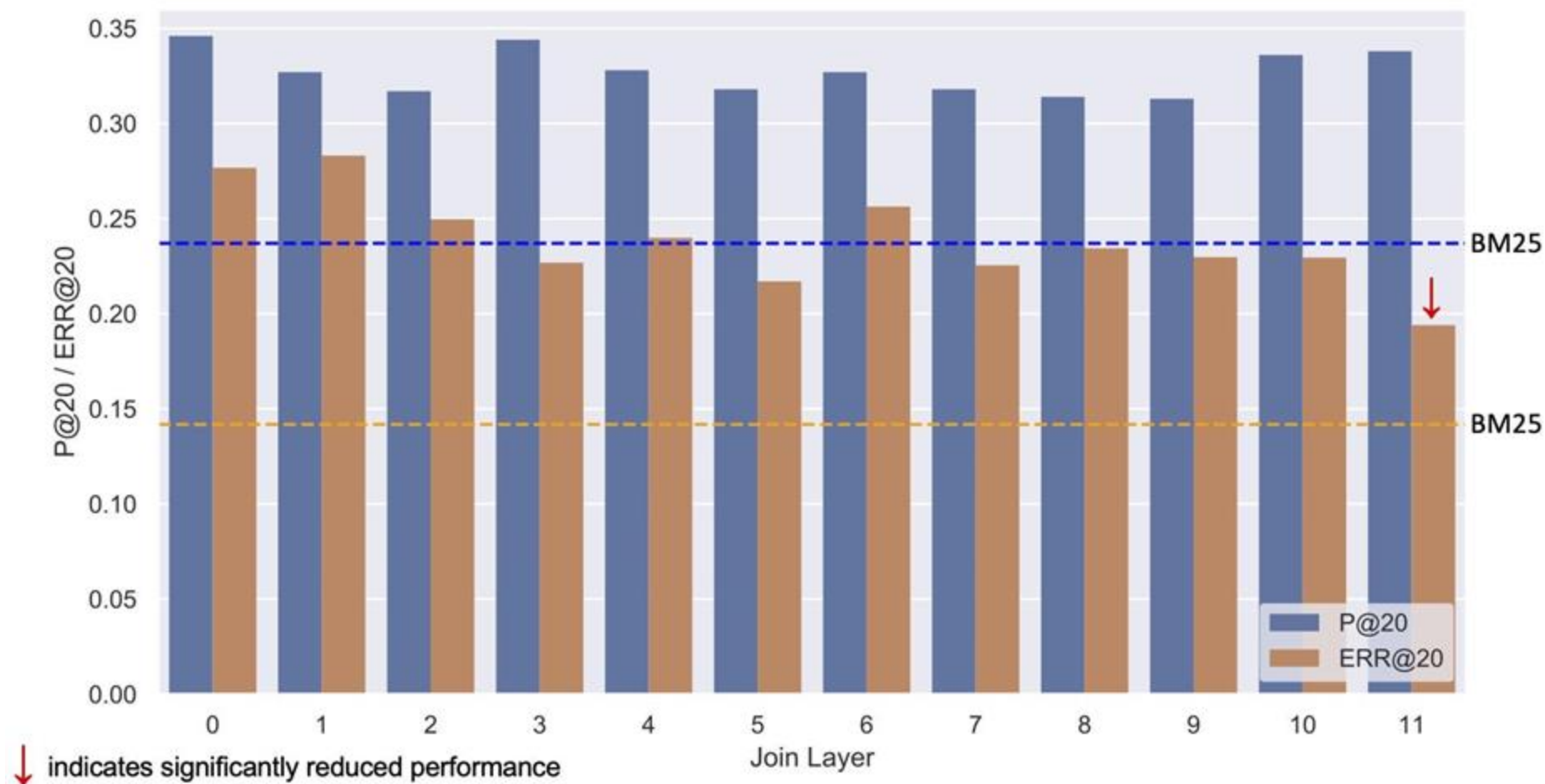


How does BERT work?



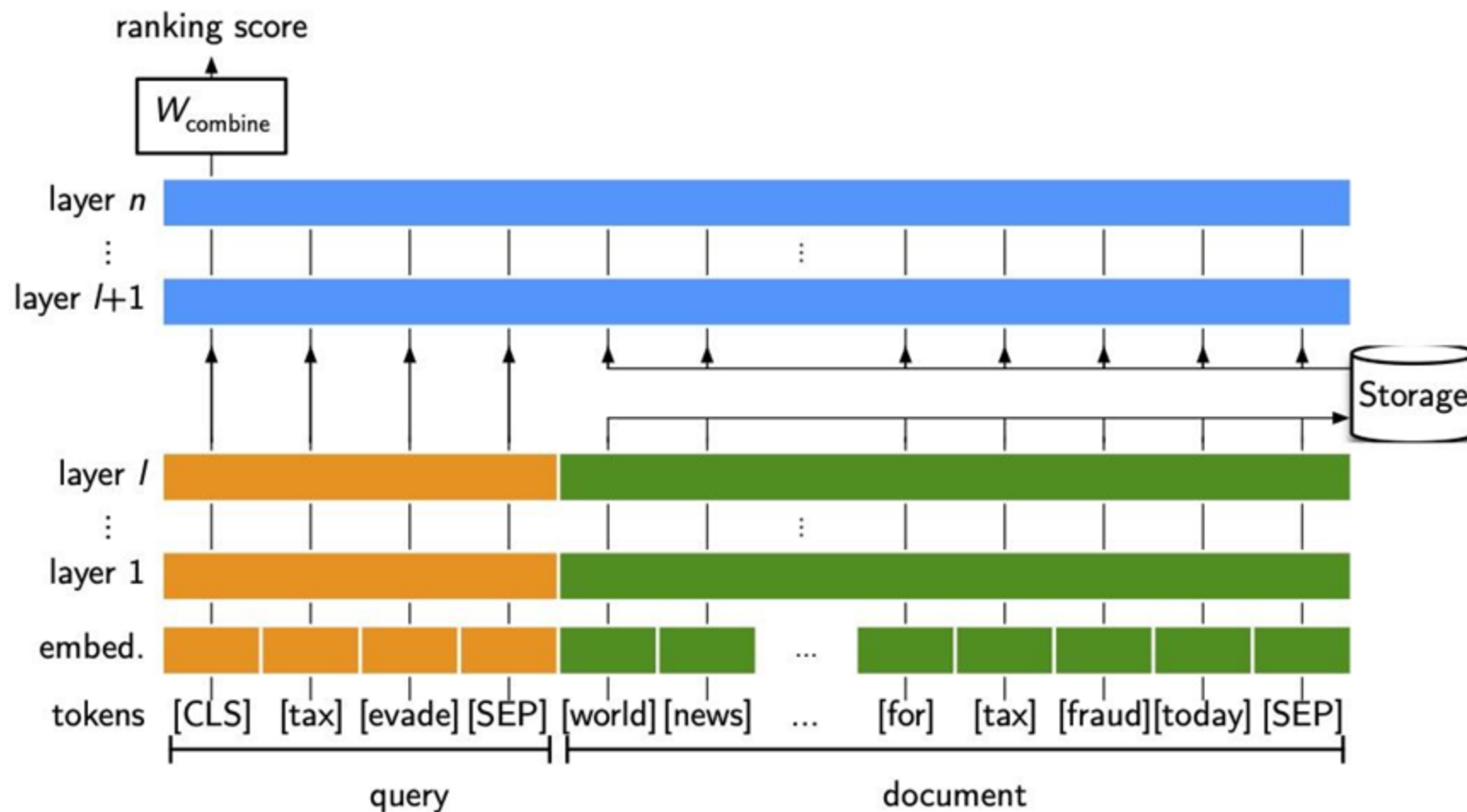
Efficient Document Re-Ranking by Precomputing Term Representations

Question: Do the early stages of self-attention between the query and document matter for relevance? (spoiler: not really)



Efficient Document Re-Ranking by Precomputing Term Representations

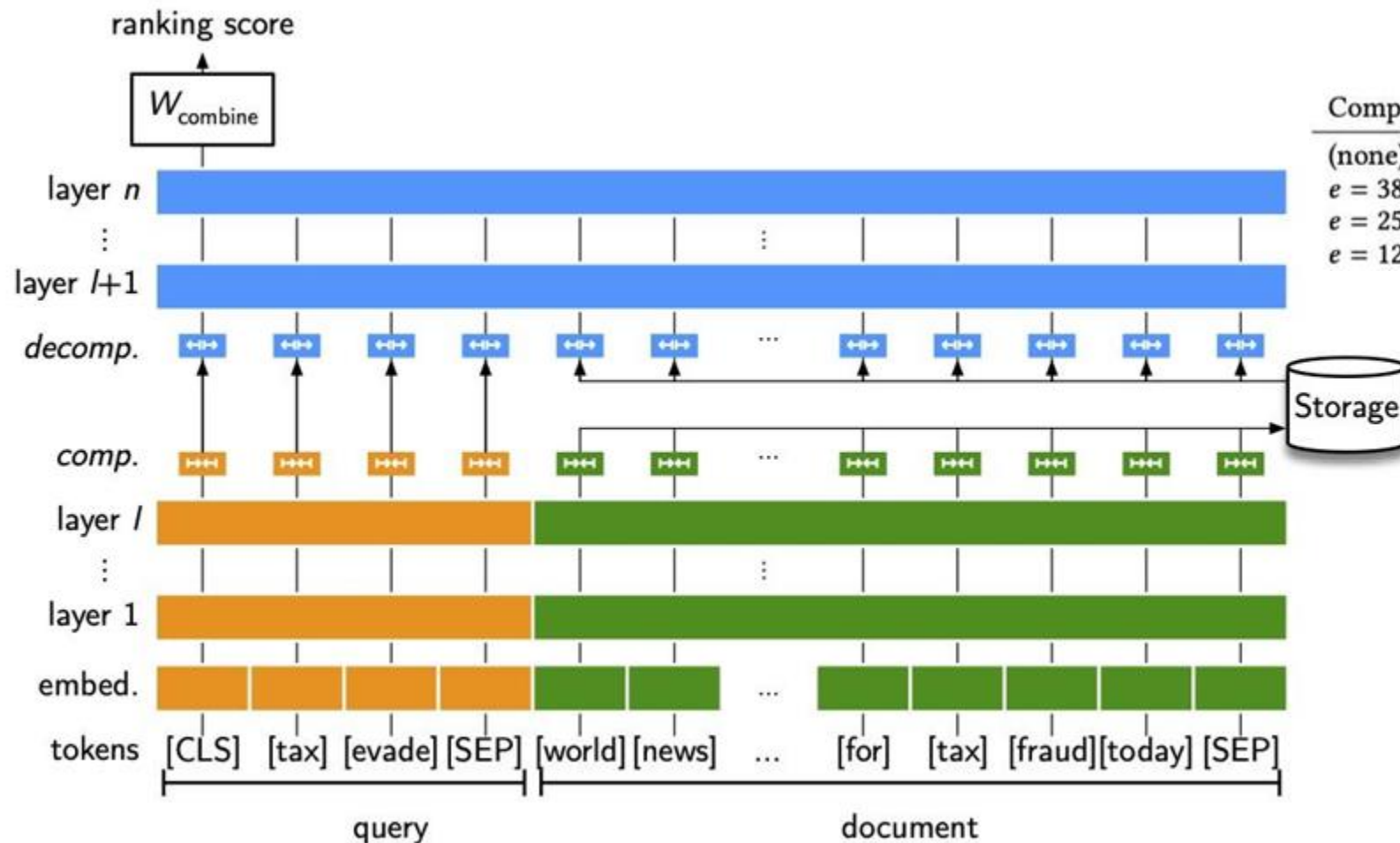
Idea: Query and document text can be processed independently in early transformer layers (with improved inference speed!!! :))



Efficient Document Re-Ranking by Precomputing Term Representations

Problem: Storage burden (solved with a compression/decompression layer).

Results: Up to 83% compression with no loss in accuracy (P@20, WebTrack)

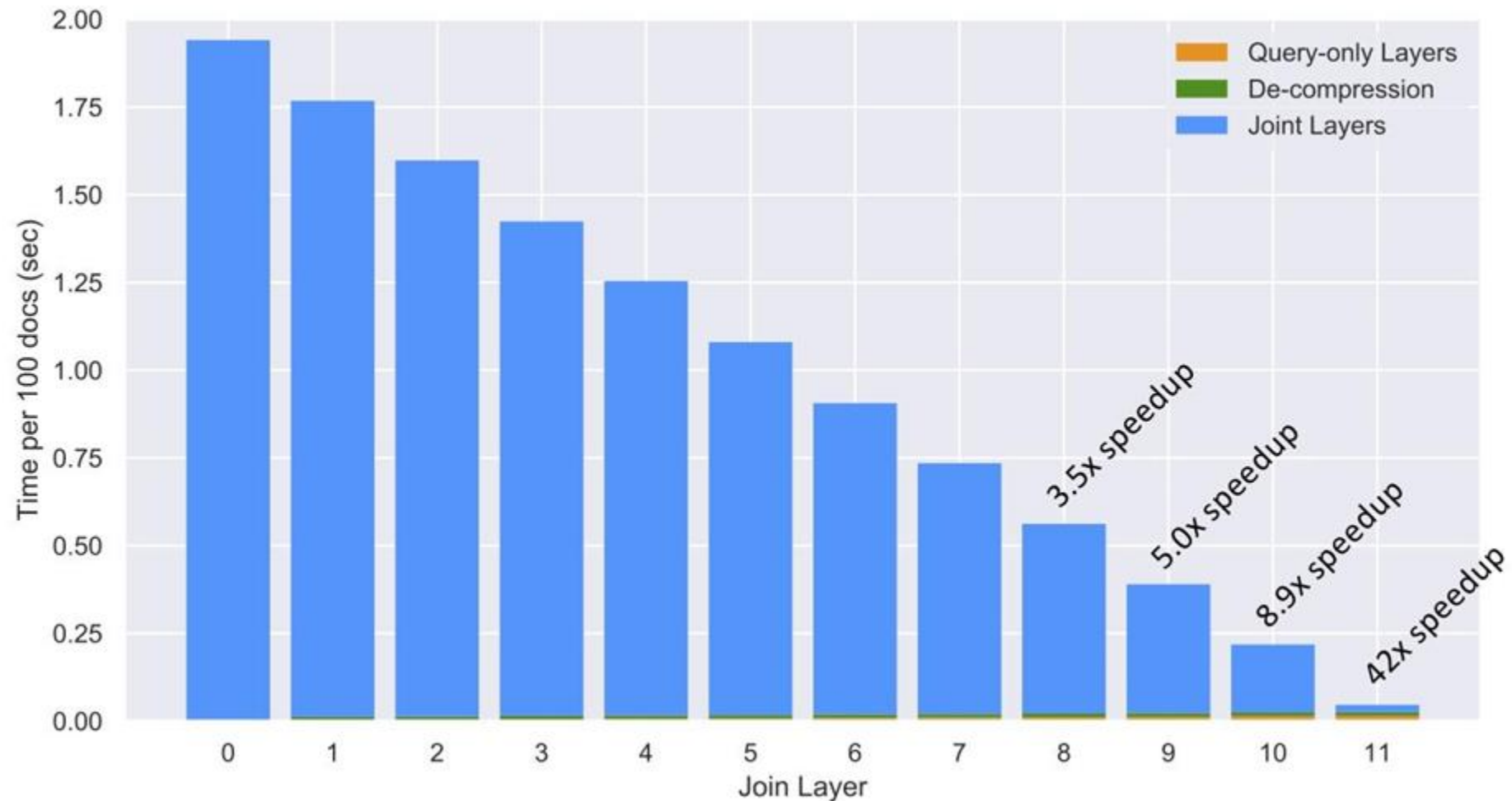


Compression	P@20				
	$l = 7$	$l = 8$	$l = 9$	$l = 10$	$l = 11$
(none)	0.3180	0.3140	0.3130	0.3360	0.3380
$e = 384$ (50%)	0.3430	0.3260	0.2980	0.3360	0.3090
$e = 256$ (67%)	0.3380	0.3120	↑ 0.3440	0.3260	0.3250
$e = 128$ (83%)	0.3100	0.3210	0.3320	0.3220	0.3370

Efficient Document Re-Ranking by Precomputing Term Representations

P@20, WebTrack2012

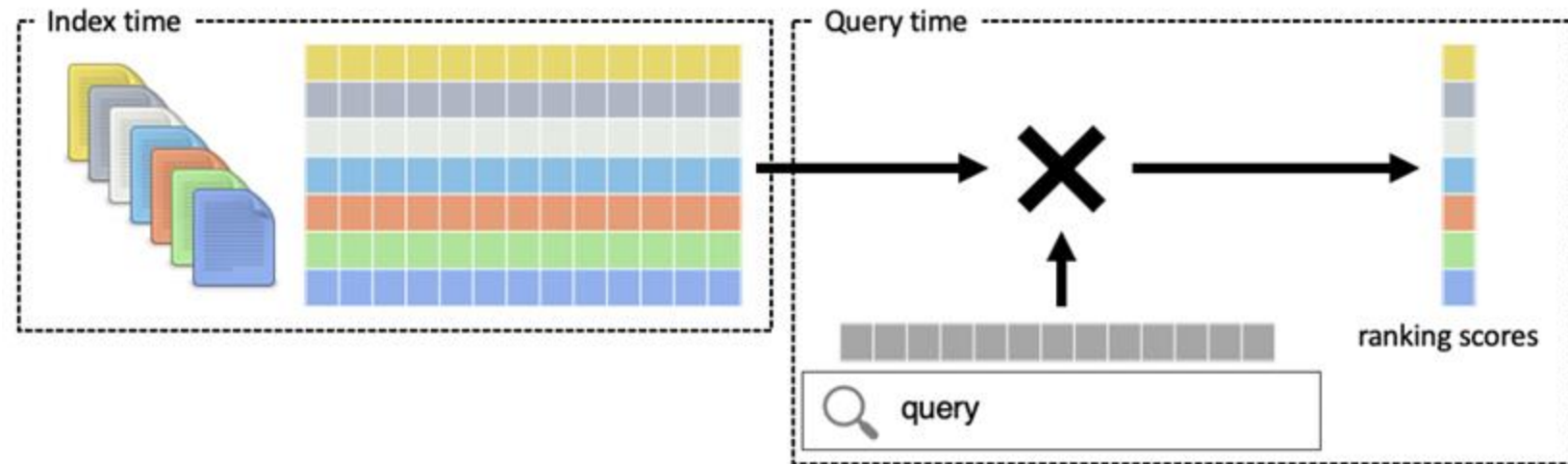
- BERT base: **0.3460** vs BERT (PreTTR, l=11, 83% compression): **0.3370**



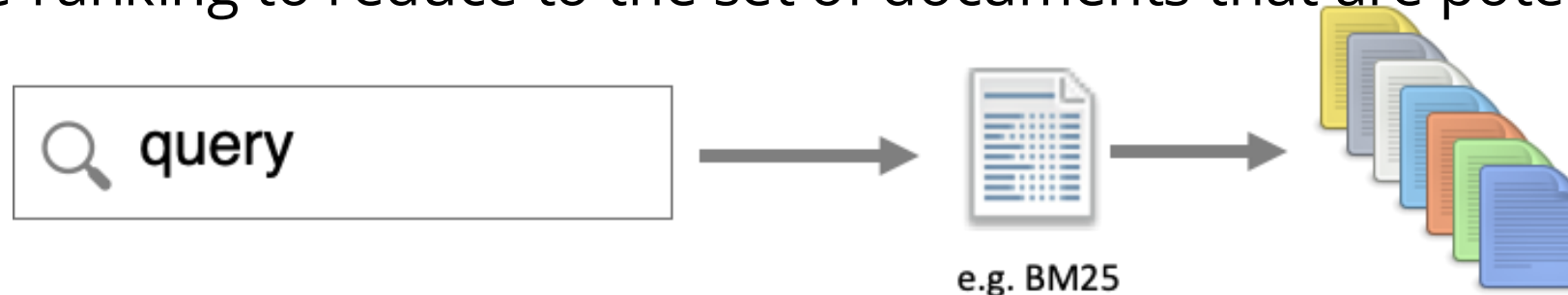
Expansion via Prediction of Importance with Contextualization (EPIC)

We want to **minimize the inference cost** while targeting **interpretability**

Idea 1: Score documents using a dot product to take advantage of efficient matrix multiplication

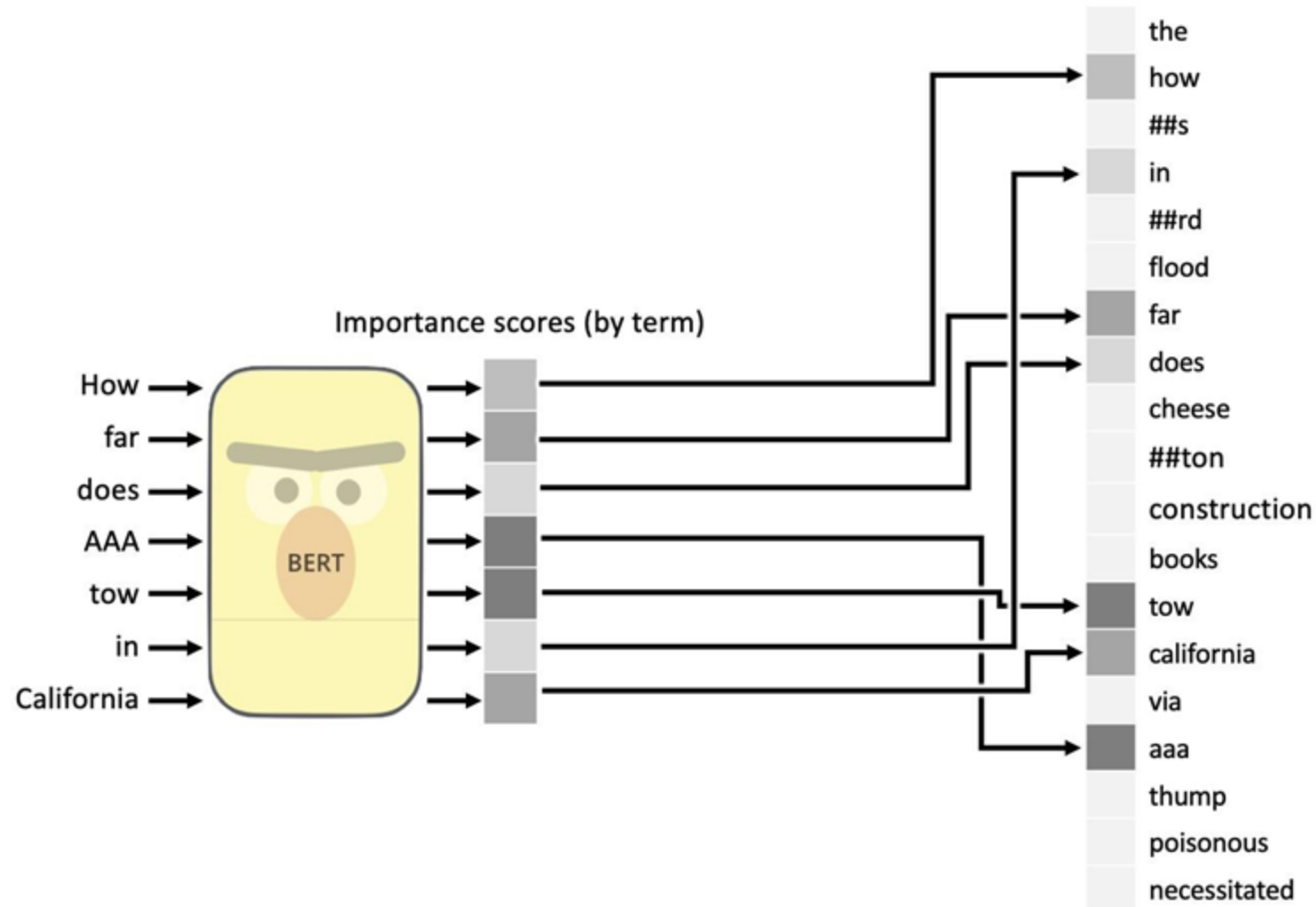


Idea 2: Use re-ranking to reduce to the set of documents that are potentially relevant



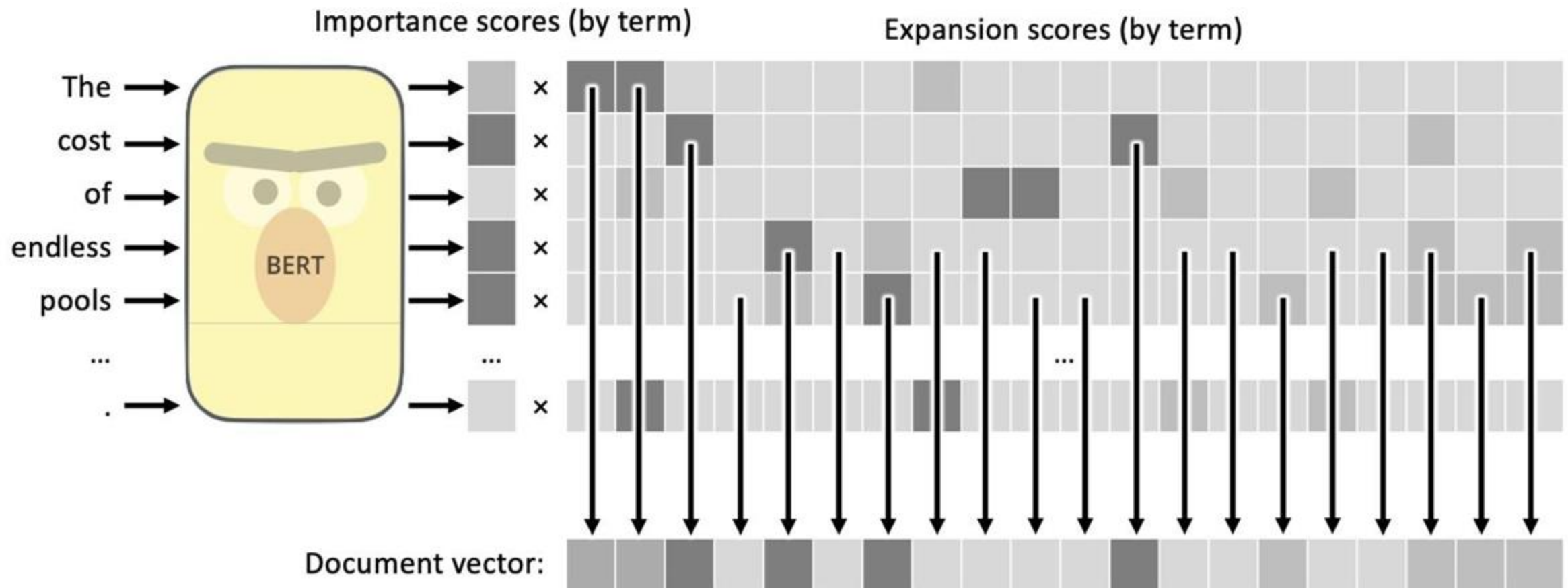
Expansion via Prediction of Importance with Contextualization (EPIC)

Query vectors: learned importance scores that directly map to positions in the representation.



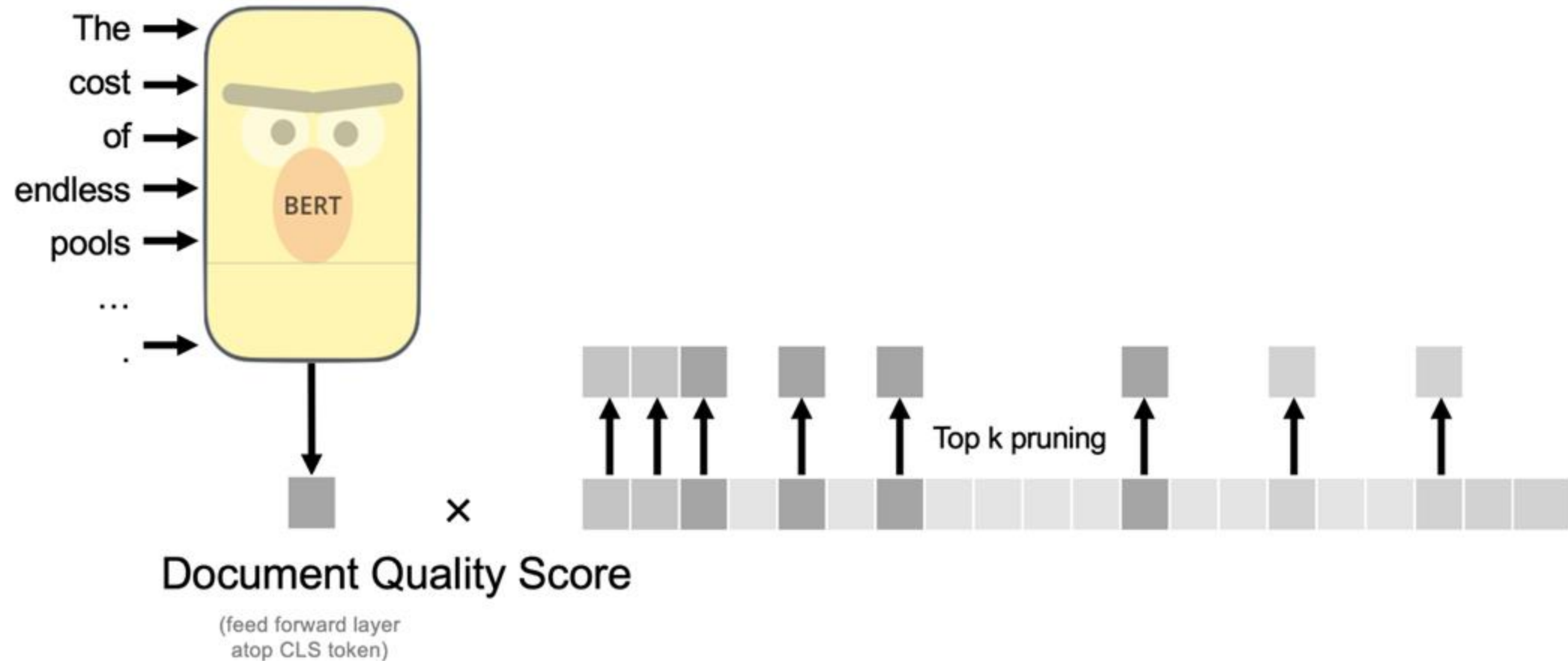
Expansion via Prediction of Importance with Contextualization (EPIC)

Document vectors: importance score plus expansion score

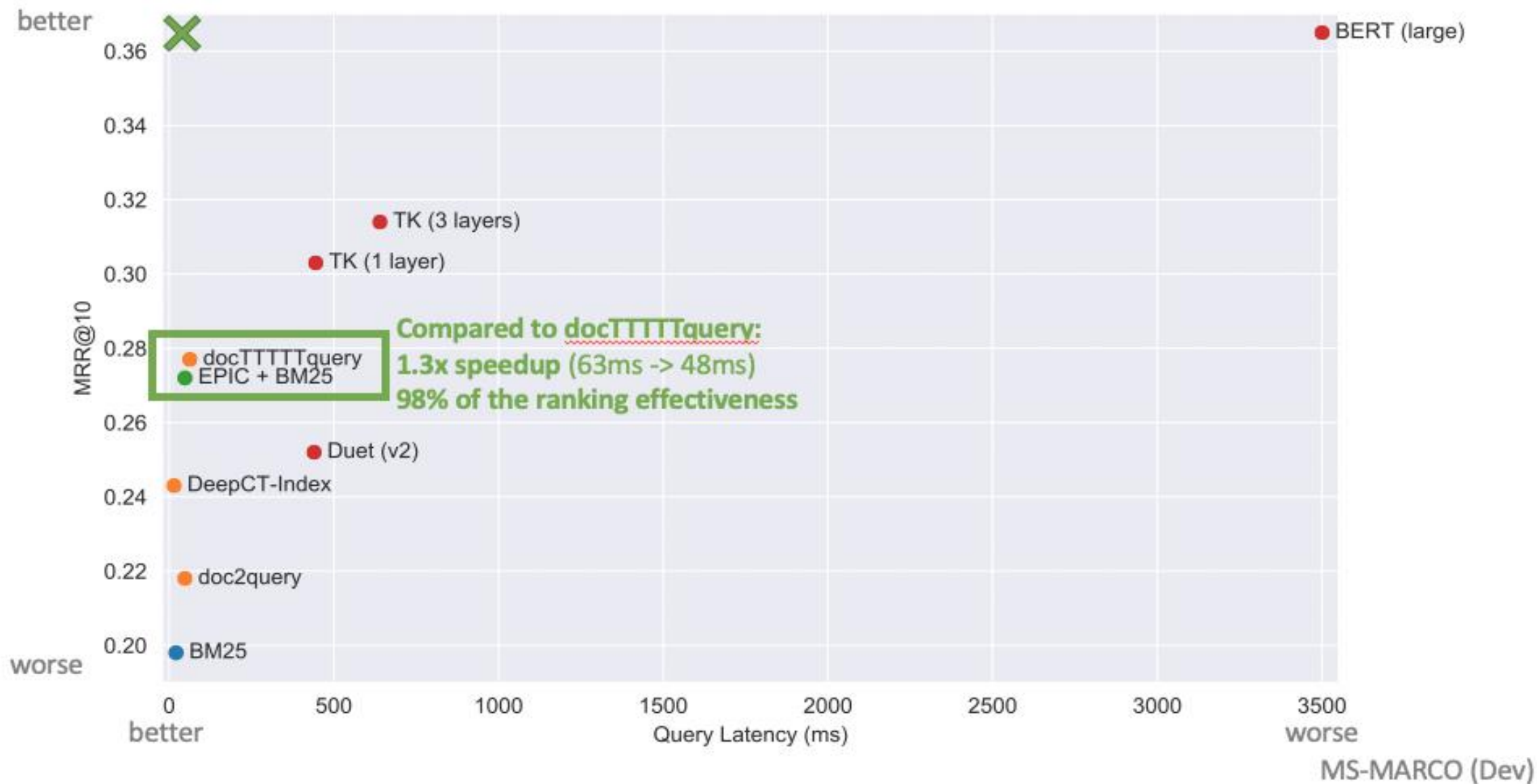


Expansion via Prediction of Importance with Contextualization (EPIC)

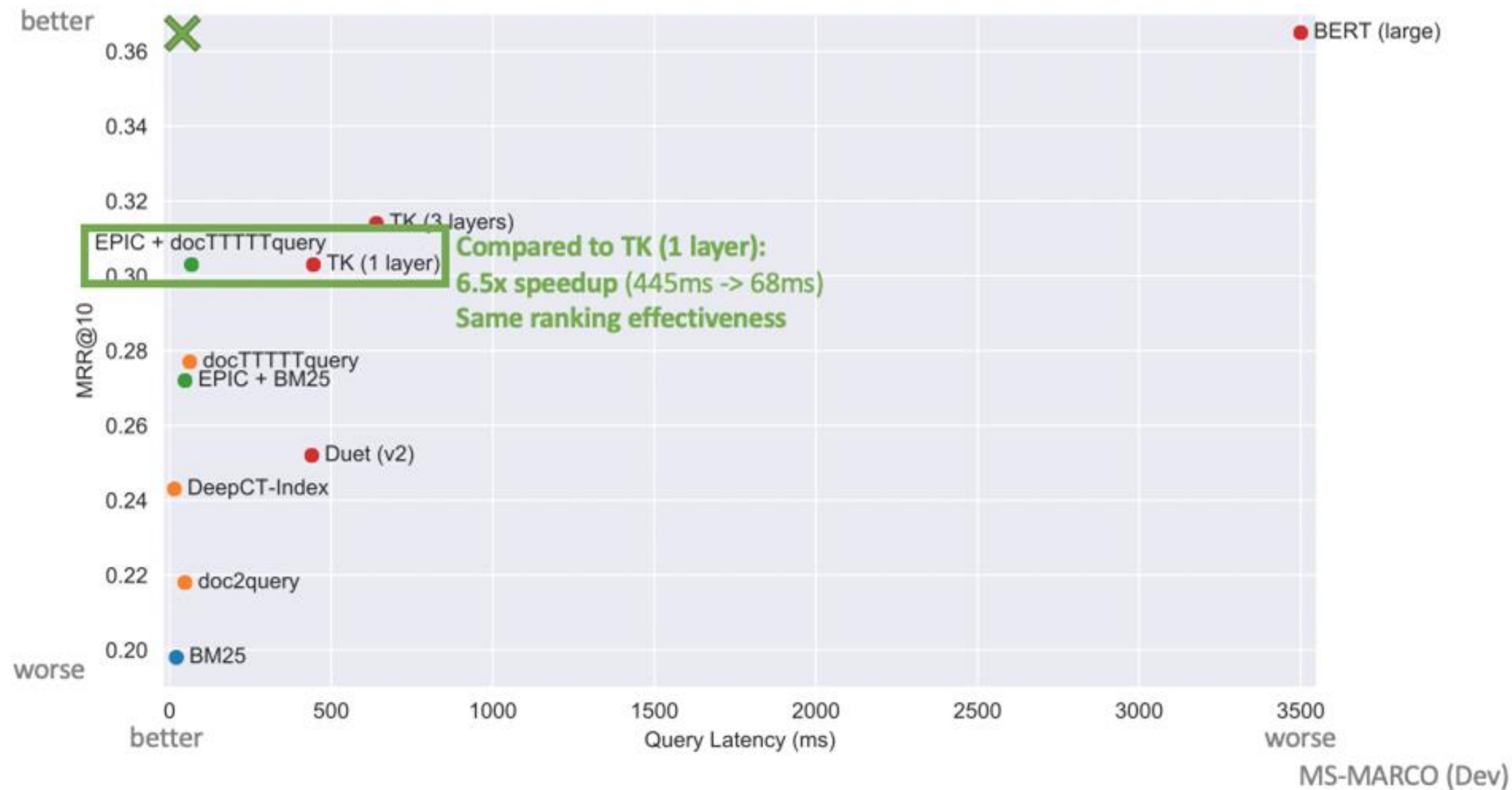
Document Quality Score + Pruning



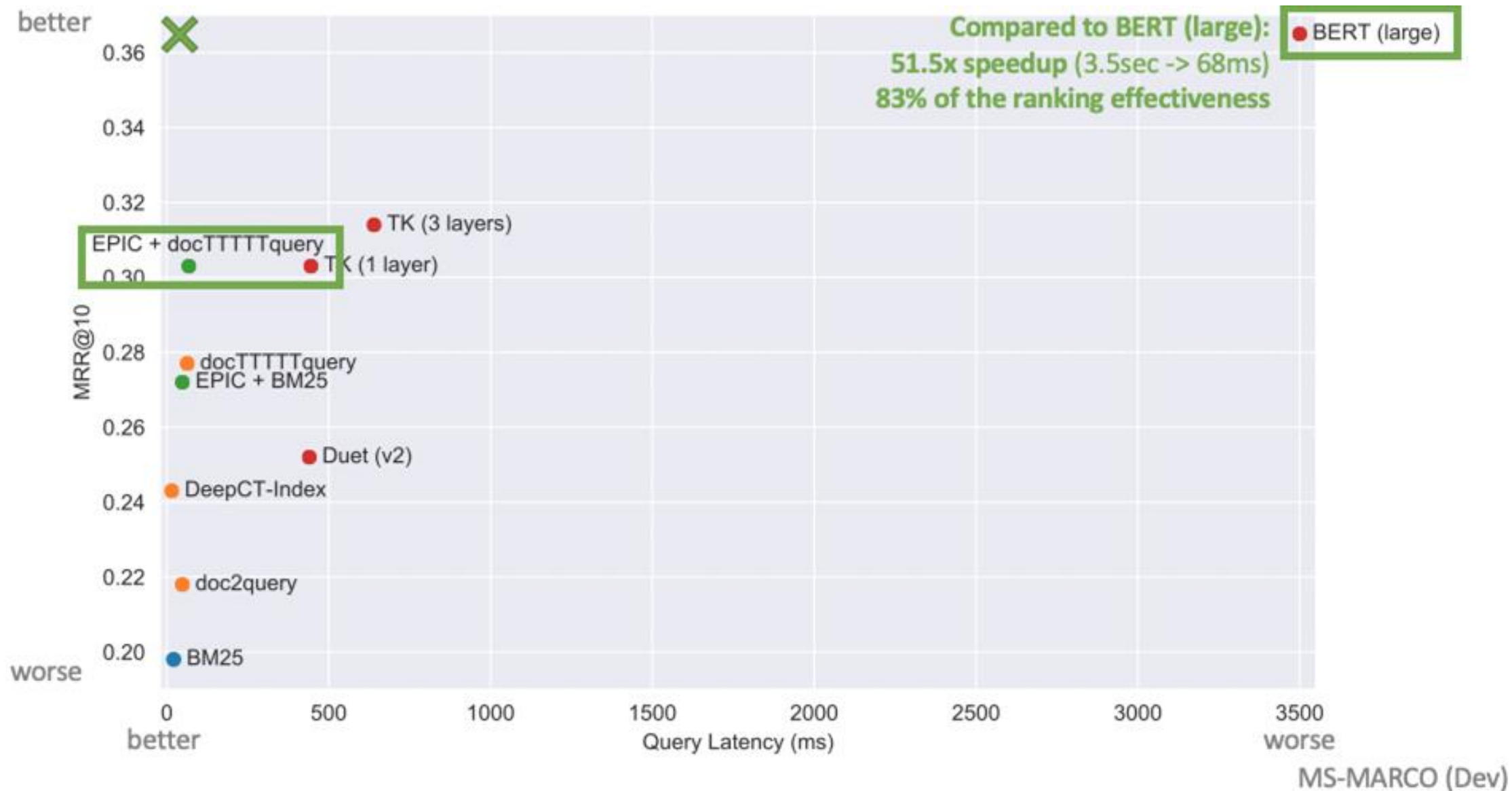
Experiments



Experiments



Experiments



EPIC allows Interpretability

(a) how far does aaa tow , in california

(b) coastal processes are located on what vertebrae

(c) cost of endless pools swim spa

Document terms: endless pools and swim spa ##s are available in a number of different price brackets . for the brand called ae ##nd ##less pools ##a prices start at \$ 23 , 900 for their most basic pool for other brands of endless pool (a . k . a swim spa ##s) prices can be as low as \$ 900 ##0 things that change the price prices change drastically depending upon location , time of year , and property type .

Top expansion terms: pay paid cost paying much what fee costs thing spending

docTTTTquery: pool endless how cost much price doe swim build spa ...